

41107 Marine and Ocean Engineering

FLOW AROUND AND FORCES ON CYLINDERS

Solved MCQ Question Bank - professional formula notation

Scope and ordering

This PDF collects all reviewed questions beginning with **FLOW AROUND AND FORCES ON CYLINDERS**, plus the directly related **Morison / Froude-Krylov / diffraction** questions that belong to the same physical block. Exact duplicates across the reviewed MCQ documents were removed. The questions are ordered as requested: first **steady current**, then **oscillatory flow / wave-related cylinder questions**, and finally **Froude-Krylov / diffraction / large-body questions**.

For each question, the structure is:

1. Original question statement and original answer options.
2. Correct answer highlighted immediately after the options.
3. **Formulario / principles used** with professional mathematical notation.
4. **Step-by-step solution** with numerical substitution whenever numerical data are present.

Default constants, unless stated otherwise

$$\rho = 1025 \text{ kg/m}^3, \quad \nu = 1.0 \times 10^{-6} \text{ m}^2/\text{s}, \quad g = 9.81 \text{ m/s}^2$$

$$A = \frac{\pi D^2}{4}, \quad \omega = \frac{2\pi}{T}$$

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Steady current

S1. Question 1

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder: The drag force can be split into a form drag and a friction drag. Which is largest when $Re > 10^4$?

Options:

- A. friction drag
- B. equal magnitude
- C. form drag

Correct answer: C. form drag **Table 2***

Formulario / principles used

- For a bluff body such as a circular cylinder at offshore-relevant Reynolds numbers, drag is mainly pressure/form drag.
- Mean drag can be decomposed conceptually as

$$F_D = F_{\text{form}} + F_{\text{friction}}.$$

- For $Re \gtrsim 10^4$, the wake and pressure difference dominate over wall friction.

Step-by-step solution

1. Identify the condition:

$$Re > 10^4.$$

No extra numerical calculation is needed because the Reynolds-number threshold is already given.

2. At this Reynolds number, separation creates a large wake behind the cylinder.
3. The main contribution to drag is therefore the pressure difference between the front and rear of the cylinder:

$$F_{\text{form}} \gg F_{\text{friction}}.$$

4. Hence the largest contribution is form drag.

Source(s): Test MC exam_Corr / Test_Exam_all

S2. Question 2

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder. How many vortices are shed during one vortex shedding period?

Options:

- A. 1
- B. 4
- C. 2

Correct answer: C. 2

Formulario / principles used

- Vortex shedding behind a free circular cylinder is alternating: one vortex from one side, then one from the other side.
- One complete shedding period contains one complete alternating cycle.

Step-by-step solution

1. During one half of the cycle, one vortex is shed from one side of the cylinder.
2. During the other half of the cycle, one vortex is shed from the opposite side.
3. Therefore, over one full vortex shedding period:

$$N_{\text{vortices}} = 1 + 1 = 2.$$

Source(s): Test MC exam_Corr / Test_Exam_all

S3. Question 3

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder: What is the pressure in excess of hydrostatic pressure in the stagnation point?

Options:

- A. $c_p = 0$
- B. $\frac{1}{2}\rho U_c^2$
- C. 1 Pa

Correct answer: B. $\frac{1}{2}\rho U_c^2$

Formulario / principles used

- Stagnation pressure increase from Bernoulli:

$$p_s - p_0 = \frac{1}{2}\rho U_c^2.$$

- At the stagnation point, the local velocity is zero and the incoming kinetic energy per unit volume becomes pressure.

Step-by-step solution

1. Apply Bernoulli between the undisturbed flow and the stagnation point:

$$p_s - p_0 = \frac{1}{2}\rho U_c^2.$$

2. No numerical value of U_c is given, so the result remains symbolic.
3. Therefore, the pressure in excess of hydrostatic pressure is

$$p_s - p_0 = \frac{1}{2}\rho U_c^2.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

S4. Question 4

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free smooth circular cylinder, the wake is turbulent, $U_c = 0.5$ m/s, and $D = 0.2$ m. Which flow regime describes the flow?

Options:

- A. Trans-critical
- B. Super-critical
- C. Sub-critical

Correct answer: C. Sub-critical **Table 1***

Formulario / principles used

- Reynolds number:

$$Re = \frac{U_c D}{\nu}.$$

- For a smooth circular cylinder, a useful course-level classification is:

$$300 \lesssim Re \lesssim 3 \times 10^5 \Rightarrow \text{sub-critical regime.}$$

Step-by-step solution

1. Insert the given values into the Reynolds number:

$$Re = \frac{U_c D}{\nu} = \frac{(0.5)(0.2)}{1.0 \times 10^{-6}}.$$

2. Compute the numerator:

$$(0.5)(0.2) = 0.10.$$

3. Divide by the kinematic viscosity:

$$Re = \frac{0.10}{1.0 \times 10^{-6}} = 1.0 \times 10^5.$$

4. Compare with the regime limits:

$$1.0 \times 10^5 < 3 \times 10^5.$$

5. Therefore, the flow is in the sub-critical regime.

Source(s): Test MC exam_Corr / Test_Exam_all

S5. Question 5

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder: Consider the following three cases defined by D and U_c . You can assume $C_D = 1.2$ in the sub-critical flow regime, 0.6 in the super-critical flow regime and 0.9 in the trans-critical flow regime. Which case has the highest drag force?

Options:

- A. $D = 0.2$ m, $U_c = 0.5$ m/s
- B. $D = 5.0$ m, $U_c = 0.2$ m/s
- C. $D = 0.5$ m, $U_c = 0.2$ m/s

Correct answer: B. $D = 5.0$ m, $U_c = 0.2$ m/s

Table 1*

Formulario / principles used

- Mean drag force per unit length:

$$F_D = \frac{1}{2} \rho C_D D U_c^2.$$

- Since ρ and $1/2$ are common to all cases, compare:

$$C_D D U_c^2.$$

- Reynolds number for identifying the regime:

$$Re = \frac{U_c D}{\nu}.$$

Step-by-step solution

1. Case A Reynolds number:

$$Re_A = \frac{(0.5)(0.2)}{1.0 \times 10^{-6}} = 1.0 \times 10^5.$$

This is sub-critical, so $C_D = 1.2$. Drag factor:

$$C_D D U_c^2 = (1.2)(0.2)(0.5)^2 = (1.2)(0.2)(0.25) = 0.060.$$

2. Case B Reynolds number:

$$Re_B = \frac{(0.2)(5.0)}{1.0 \times 10^{-6}} = 1.0 \times 10^6.$$

This is super-critical, so $C_D = 0.6$. Drag factor:

$$C_D D U_c^2 = (0.6)(5.0)(0.2)^2 = (0.6)(5.0)(0.04) = 0.120.$$

3. Case C Reynolds number:

$$Re_C = \frac{(0.2)(0.5)}{1.0 \times 10^{-6}} = 1.0 \times 10^5.$$

This is sub-critical, so $C_D = 1.2$. Drag factor:

$$C_D D U_c^2 = (1.2)(0.5)(0.2)^2 = (1.2)(0.5)(0.04) = 0.024.$$

4. Compare:

$$0.120 > 0.060 > 0.024.$$

5. Therefore, Case B has the highest drag force.

Source(s): MCQ Test 2025 / Test_Exam_all

S6. Question 6

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free smooth circular cylinder, the wake is turbulent and the vortex shedding frequency, $f_v = 0.018$ Hz, $U_c = 0.2$ m/s, and $D = 5$ m. Which flow regime describes the flow?

Options:

- A. Trans-critical
- B. Super-critical
- C. Sub-critical

Correct answer: B. Super-critical

Table 1*

Formulario / principles used

- Reynolds number:

$$Re = \frac{U_c D}{\nu}.$$

- Strouhal number:

$$St = \frac{f_v D}{U_c}.$$

- For a smooth cylinder, $St \approx 0.45$ is typical of the super-critical regime.

Step-by-step solution

1. Compute the Reynolds number:

$$Re = \frac{(0.2)(5)}{1.0 \times 10^{-6}} = \frac{1.0}{1.0 \times 10^{-6}} = 1.0 \times 10^6.$$

2. Compute the Strouhal number:

$$St = \frac{f_v D}{U_c} = \frac{(0.018)(5)}{0.2}.$$

3. Numerical substitution:

$$(0.018)(5) = 0.090, \quad St = \frac{0.090}{0.2} = 0.45.$$

4. A Strouhal number around 0.45, together with $Re = 10^6$, indicates the super-critical regime.

Source(s): MCQ Test 2025 / Test_Exam_all

S7. Question 7

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. f_v is the vortex shedding frequency. What is the frequency of the oscillating drag force?

Options:

- A. $\frac{1}{2}f_v$
- B. f_v
- C. $2f_v$

Correct answer: C. $2f_v$

Formulario / principles used

- Oscillating drag-force frequency:

$$f_{D,osc} = 2f_v.$$

- Physical reason: drag receives two similar pulses per full vortex shedding cycle, while lift changes sign once per alternating cycle.

Step-by-step solution

- The formula is directly:

$$f_{D,osc} = 2f_v.$$

- No numerical value of f_v is given, so the result remains symbolic.

- Therefore, the oscillating drag force has frequency $2f_v$.

Source(s): MCQ Test 2025 / Test_Exam_all

S8. Question 8

FLOW AROUND AND FORCES ON CYLINDERS. Steady current around a free smooth circular cylinder. Consider a cylinder with $D = 1.0$ m in a current $U_c = 0.25$ m/s. Take $\nu = 1.0 \times 10^{-6}$ m²/s. Which flow regime is most appropriate?

Options:

- Sub-critical
- Critical / trans-critical onset
- Super-critical
- No vortex shedding because $Re < 1$

Correct answer: A. Sub-critical Table 1*

Formulario / principles used

- Reynolds number:

$$Re = \frac{U_c D}{\nu}.$$

- Smooth-cylinder sub-critical range, approximately:

$$300 \lesssim Re \lesssim 3 \times 10^5.$$

Step-by-step solution

1. Substitute the data:

$$Re = \frac{(0.25)(1.0)}{1.0 \times 10^{-6}}.$$

2. Compute:

$$Re = \frac{0.25}{1.0 \times 10^{-6}} = 2.5 \times 10^5.$$

3. Compare with the threshold:

$$2.5 \times 10^5 < 3.0 \times 10^5.$$

4. The value is close to the transition region, but it is still below the usual critical onset.
5. Therefore, the most appropriate answer is sub-critical.

Source(s): 41107_mock_exam_with_answers

S9. Question 9

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. Consider the following three cases. Assume $C_D = 1.2$ in the sub-critical regime and $C_D = 0.9$ in the trans-critical regime. Which case gives the largest mean drag force per unit length?

Options:

- A. $D = 0.3$ m, $U_c = 0.4$ m/s
- B. $D = 4.0$ m, $U_c = 0.25$ m/s
- C. $D = 0.8$ m, $U_c = 0.6$ m/s

Correct answer: C. $D = 0.8$ m, $U_c = 0.6$ m/s **Table 1***

Formulario / principles used

- Mean drag force per unit length:

$$F_D = \frac{1}{2} \rho C_D D U_c^2.$$

- Compare only the proportional factor:

$$F_D \propto C_D D U_c^2.$$

Step-by-step solution

1. Case A:

$$Re_A = \frac{(0.4)(0.3)}{1.0 \times 10^{-6}} = 1.2 \times 10^5.$$

Use $C_D = 1.2$. Then:

$$C_D DU_c^2 = (1.2)(0.3)(0.4)^2 = (1.2)(0.3)(0.16) = 0.0576.$$

2. Case B:

$$Re_B = \frac{(0.25)(4.0)}{1.0 \times 10^{-6}} = 1.0 \times 10^6.$$

Using the high-Re coefficient specified by the question, $C_D = 0.9$:

$$C_D DU_c^2 = (0.9)(4.0)(0.25)^2 = (0.9)(4.0)(0.0625) = 0.225.$$

3. Case C:

$$Re_C = \frac{(0.6)(0.8)}{1.0 \times 10^{-6}} = 4.8 \times 10^5.$$

Using the high-Re coefficient specified by the question, $C_D = 0.9$:

$$C_D DU_c^2 = (0.9)(0.8)(0.6)^2 = (0.9)(0.8)(0.36) = 0.2592.$$

4. Compare the three drag factors:

$$0.2592 > 0.225 > 0.0576.$$

5. Therefore, Case C gives the largest drag force.

Source(s): 41107_mock_exam_with_answers

S10. Question 10

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free cylinder. What is the pressure in excess of hydrostatic pressure at the stagnation point if $U_c = 1.6$ m/s and $\rho = 1025$ kg/m³?

Options:

- A. 0 Pa
- B. 820 Pa
- C. 1310 Pa
- D. 2620 Pa

Correct answer: C. 1310 Pa

Formulario / principles used

- Stagnation pressure increase:

$$p_s - p_0 = \frac{1}{2} \rho U_c^2.$$

Step-by-step solution

1. Substitute the data:

$$p_s - p_0 = \frac{1}{2}(1025)(1.6)^2.$$

2. Square the velocity:

$$(1.6)^2 = 2.56.$$

3. Multiply:

$$p_s - p_0 = 512.5 \times 2.56 = 1312 \text{ Pa.}$$

4. Round to the closest option:

$$p_s - p_0 \approx 1310 \text{ Pa.}$$

Source(s): 41107_mock_exam_with_answers

S11. Question 11

FLOW AROUND AND FORCES ON CYLINDERS. Steady flow around a free circular cylinder. If the vortex shedding frequency is $f_v = 0.24 \text{ Hz}$, what is the frequency of the oscillating drag force?

Options:

- A. 0.12 Hz
- B. 0.24 Hz
- C. 0.48 Hz
- D. 0.96 Hz

Correct answer: C. 0.48 Hz

Formulario / principles used

- Oscillating drag-force frequency:

$$f_{D,\text{osc}} = 2f_v.$$

Step-by-step solution

1. Substitute the vortex shedding frequency:

$$f_{D,\text{osc}} = 2(0.24 \text{ Hz}).$$

2. Compute:

$$f_{D,\text{osc}} = 0.48 \text{ Hz.}$$

Source(s): 41107_mock_exam_with_answers

S12. Question 12

FLOW AROUND AND FORCES ON CYLINDERS. For a smooth circular cylinder in the sub-critical regime, why is C_D almost constant over a wide range of Re ?

Options:

- A. Because separation occurs early and at nearly the same angular position, so the wake width stays similar.

- B. Because friction drag dominates completely and pressure drag is negligible.
- C. Because the boundary layer is turbulent from the stagnation point.
- D. Because vortex shedding stops in the sub-critical regime.

Correct answer: A. Because separation occurs early and at nearly the same angular position, so the wake width stays similar.

Formulario / principles used

- Drag coefficient:

$$C_D = \frac{F_D}{\frac{1}{2}\rho U_c^2 D}.$$

- In the sub-critical regime, the boundary layer is laminar up to separation, and the separation point stays relatively stable.

Step-by-step solution

1. In the sub-critical regime, separation occurs early on the cylinder.
2. Because the separation point does not move dramatically with Re , the wake width remains similar.
3. Since form drag is controlled mainly by wake width and pressure difference, C_D remains approximately constant.
4. No numerical substitution is required because this is a conceptual regime question.

Source(s): 41107_mock_exam_with_answers

Oscillatory flow / wave-related FLOW-labelled questions

O1. Question 1

FLOW AROUND AND FORCES ON CYLINDERS. The wave period is T . What is the definition of the KC number?

Options:

- A. $KC = \frac{U_m T}{D}$
- B. $KC = \frac{U_m D}{T}$
- C. $KC = \frac{T D}{U_m}$

Correct answer: A. $KC = \frac{U_m T}{D}$

Formulario / principles used

- Keulegan-Carpenter number:

$$KC = \frac{U_m T}{D}.$$

- $U_m T$ has units of length and represents a characteristic fluid excursion over one period.

Step-by-step solution

- Start from the definition:

$$KC = \frac{\text{fluid excursion length}}{\text{cylinder diameter}}.$$

- The characteristic excursion length is:

$$\text{fluid excursion} \sim U_m T.$$

- Therefore:

$$KC = \frac{U_m T}{D}.$$

- No numerical substitution is needed because the question asks for the definition.

Source(s): Test MC exam_Corr

O2. Question 2

FLOW AROUND AND FORCES ON CYLINDERS. The wave period is T . What is the definition of the KC number?

Options:

- A. $KC = \frac{U_m D}{\nu}$
- B. $KC = \frac{0.44}{D/L}$
- C. $KC = \frac{U_m T}{D}$

Correct answer: C. $KC = \frac{U_m T}{D}$

Formulario / principles used

- Keulegan-Carpenter number:

$$KC = \frac{U_m T}{D}.$$

- Do not confuse it with wave Reynolds number:

$$Re = \frac{U_m D}{\nu}.$$

Step-by-step solution

1. Option A, $U_m D / \nu$, is the wave Reynolds number, not KC .
2. Option B is not the definition of KC .
3. Option C matches the correct definition:

$$KC = \frac{U_m T}{D}.$$

4. No numerical substitution is needed because the question asks for the definition.

Source(s): MCQ Test 2025 / Test_Exam_all

O3. Question 3

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free, smooth circular cylinder: $D = 1$ m, $T = 10$ s, $U_m = 0.4$ m/s. What is the maximum value of the Froude-Krylov force?

Options:

- A. 396 N/m
- B. 193 N/m
- C. 202 N/m

Correct answer: C. 202 N/m

Formulario / principles used

- Froude-Krylov force per unit length:

$$F_{FK} = \rho A \dot{U}.$$

- For $U(t) = U_m \sin(\omega t)$, the maximum acceleration is:

$$\dot{U}_{\max} = U_m \omega.$$

- Therefore:

$$F_{FK,\max} = \rho A U_m \omega.$$

- Circular cylinder cross-sectional area:

$$A = \frac{\pi D^2}{4}.$$

- Angular frequency:

$$\omega = \frac{2\pi}{T}.$$

Step-by-step solution

1. Compute the cross-sectional area:

$$A = \frac{\pi D^2}{4} = \frac{\pi (1)^2}{4} = 0.7854 \text{ m}^2.$$

2. Compute the angular frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{10} = 0.6283 \text{ rad/s}.$$

3. Substitute into the maximum Froude-Krylov force:

$$F_{FK,\max} = \rho A U_m \omega = (1025)(0.7854)(0.4)(0.6283).$$

4. Numerical multiplication:

$$F_{FK,\max} = 202.3 \text{ N/m}.$$

5. Closest option:

$$F_{FK,\max} \approx 202 \text{ N/m}.$$

Source(s): Test MC exam_Corr / Test_Exam_all

O4. Question 4

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free, smooth circular cylinder. Consider the case defined by $U_m = 1 \text{ m/s}$, $D = 1 \text{ m}$ and the wave period $T = 5 \text{ s}$. Further, assume the following functional expression for determining C_D and C_M . $C_D = 1.3 + 0.1(KC - 12)$ and $C_M = \min(2.0; 2.0 - 0.044(KC - 3))$ for $2 < KC < 12$. What is the ratio between the maximum inertia and maximum drag force ($F_{I,\max}/F_{D,\max}$)?

Options:

- A. 0.16
- B. 4.00
- C. 6.29

Correct answer: C. 6.29

Formulario / principles used

- Keulegan-Carpenter number:

$$KC = \frac{U_m T}{D}.$$

- Maximum inertia-to-drag ratio:

$$\frac{F_{I,\max}}{F_{D,\max}} = \frac{\pi^2}{KC} \frac{C_M}{C_D}.$$

- Coefficients from the question:

$$C_D = 1.3 + 0.1(KC - 12), \quad C_M = \min(2.0, 2.0 - 0.044(KC - 3)).$$

Step-by-step solution

1. Compute KC :

$$KC = \frac{(1)(5)}{1} = 5.$$

2. Compute C_D :

$$C_D = 1.3 + 0.1(5 - 12) = 1.3 + 0.1(-7) = 1.3 - 0.7 = 0.6.$$

3. Compute C_M :

$$C_M = \min(2.0, 2.0 - 0.044(5 - 3)).$$

$$C_M = \min(2.0, 2.0 - 0.088) = \min(2.0, 1.912) = 1.912.$$

4. Substitute into the force ratio:

$$\frac{F_{I,\max}}{F_{D,\max}} = \frac{\pi^2}{5} \frac{1.912}{0.6}.$$

5. Numerical calculation:

$$\frac{\pi^2}{5} = \frac{9.8696}{5} = 1.9739, \quad \frac{1.912}{0.6} = 3.1867.$$

$$\frac{F_{I,\max}}{F_{D,\max}} = (1.9739)(3.1867) = 6.29.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

O5. Question 5

MORISON EQUATION - Froude-Krylov vs Inertia Force. What is the relationship between the Froude-Krylov force F_K and the total inertia force F_I in the Morison equation?

Options:

- A. $F_I = C_M F_K$ (the Froude-Krylov is multiplied by C_M)
- B. $F_I = F_K / C_M$

C. $F_I = F_K$ + extra diffraction force

Correct answer: A. $F_I = C_M F_K$

Formulario / principles used

- Morison equation for a stationary cylinder in oscillatory flow:

$$F(t) = \frac{1}{2} \rho C_D D U |U| + \rho C_M A \ddot{U}.$$

- Froude-Krylov force:

$$F_K = \rho A \ddot{U}.$$

- Total inertia force:

$$F_I = \rho C_M A \ddot{U}.$$

- Coefficient relation:

$$C_M = C_m + 1.$$

Step-by-step solution

1. Start from the Froude-Krylov force:

$$F_K = \rho A \ddot{U}.$$

2. Start from the inertia term in the Morison equation:

$$F_I = \rho C_M A \ddot{U}.$$

3. Factor out the Froude-Krylov part:

$$F_I = C_M (\rho A \ddot{U}).$$

4. Since $\rho A \ddot{U} = F_K$:

$$F_I = C_M F_K.$$

5. Therefore, the total inertia force is the Froude-Krylov force multiplied by C_M .

Source(s): 41107_MCQ_Study_Guide

O6. Question 6

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free circular cylinder. $KC < 2$ and $Re \gg 10^3$ therefore the flow around the cylinder can be described as

Options:

- A. A pair of symmetric vortices
- B. No separation
- C. Vortex shedding

Correct answer: B. No separation Table 3*

Formulario / principles used

- In oscillatory flow, the Keulegan-Carpenter number controls the fluid excursion relative to the diameter:

$$KC = \frac{U_m T}{D}.$$

- Very small KC means the fluid does not travel far enough around the cylinder for separation and vortex shedding to develop.

Step-by-step solution

- The key given condition is:

$$KC < 2.$$

- This means the fluid excursion is less than about two diameters.
- Even though $Re \gg 10^3$, the oscillatory excursion is too small for a developed wake to form.
- Therefore, the best description is:

No separation.

- No numerical substitution is needed because the question gives the regime directly through $KC < 2$.

Source(s): Test MC exam_Corr / Test_Exam_all

O7. Question 7

FLOW AROUND AND FORCES ON CYLINDERS. A circular cylinder is placed on the seabed. $D = 0.5$ m, water depth $h = 12$ m, wave height $H = 1$ m and wave period $T = 7$ s. What is the KC number?

Options:

- A. $KC = 6.3$
- B. $KC = 4.2$
- C. $KC = 5.5$

Correct answer: B. $KC = 4.2$

Formulario / principles used

- Keulegan-Carpenter number:

$$KC = \frac{U_m T}{D}.$$

- For a cylinder at the seabed, use the maximum horizontal orbital velocity at $z = -h$:

$$U_m = \frac{\pi H}{T \sinh(kh)}.$$

- Wave dispersion relation:

$$\omega^2 = gk \tanh(kh), \quad \omega = \frac{2\pi}{T}.$$

Step-by-step solution

1. Compute angular frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{7} = 0.8976 \text{ rad/s.}$$

2. Solve the dispersion relation:

$$\omega^2 = gk \tanh(kh).$$

Numerically:

$$(0.8976)^2 = 9.81k \tanh(12k).$$

This gives:

$$k \approx 0.09897 \text{ m}^{-1}.$$

3. Compute kh :

$$kh = (0.09897)(12) = 1.1876.$$

4. Compute the hyperbolic sine term:

$$\sinh(kh) = \sinh(1.1876) \approx 1.489.$$

5. Compute maximum seabed orbital velocity:

$$U_m = \frac{\pi H}{T \sinh(kh)} = \frac{\pi(1)}{(7)(1.489)}.$$

$$U_m = \frac{3.1416}{10.423} = 0.3015 \text{ m/s.}$$

6. Compute KC :

$$KC = \frac{U_m T}{D} = \frac{(0.3015)(7)}{0.5} = 4.22.$$

7. Closest answer:

$$KC \approx 4.2.$$

Source(s): MCQ Test 2025 / Test_Exam_all

O8. Question 8

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. The normalized fundamental lift frequency is $N_L = 3$. Which name is given to this flow regime?

Options:

- A. three pairs, vortex shedding regime
- B. double pair, vortex shedding regime
- C. single pair, vortex shedding regime

Correct answer: B. double pair, vortex shedding regime Table 4*

Formulario / principles used

- Normalized fundamental lift frequency:

$$N_L = \frac{f_L}{f_w}.$$

- Course convention for oscillatory-flow regimes:

$$N_L \approx 2 \Rightarrow \text{single-pair regime,}$$

$$N_L \approx 3 \Rightarrow \text{double-pair regime,}$$

$$N_L \approx 4 \Rightarrow \text{three-pair regime.}$$

Step-by-step solution

- Given:

$$N_L = 3.$$

- Match this value with the regime map:

$$N_L = 3 \Rightarrow \text{double-pair vortex shedding regime.}$$

- No further numerical calculation is needed.

Source(s): MCQ Test 2025 / Test_Exam_all

O9. Question 9

FLOW AROUND AND FORCES ON CYLINDERS. What is the maximum wave height on a horizontal bed given a wave period $T = 7$ s and a water depth $h = 19$ m?

Options:

- A. $H_{\max} = 15.2$ m
- B. $H_{\max} = 9.3$ m
- C. $H_{\max} = 4.1$ m

Correct answer: B. $H_{\max} = 9.3$ m

Formulario / principles used

- Wave angular frequency:

$$\omega = \frac{2\pi}{T}.$$

- Linear dispersion relation:

$$\omega^2 = gk \tanh(kh).$$

- Wavelength:

$$L = \frac{2\pi}{k}.$$

- Miche-type breaking limit on a horizontal bed:

$$\frac{H_{\max}}{L} \approx 0.142 \tanh(kh),$$

so

$$H_{\max} \approx 0.142L \tanh(kh).$$

Step-by-step solution

1. Compute angular frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{7} = 0.8976 \text{ rad/s.}$$

2. Solve dispersion:

$$(0.8976)^2 = 9.81k \tanh(19k).$$

This gives:

$$k \approx 0.0881 \text{ m}^{-1}.$$

3. Compute wavelength:

$$L = \frac{2\pi}{k} = \frac{2\pi}{0.0881} = 71.3 \text{ m.}$$

4. Compute kh :

$$kh = (0.0881)(19) = 1.674.$$

5. Compute $\tanh(kh)$:

$$\tanh(1.674) = 0.932.$$

6. Insert into the breaking limit:

$$H_{\max} = 0.142L \tanh(kh) = 0.142(71.3)(0.932).$$

7. Numerical result:

$$H_{\max} = 9.44 \text{ m.}$$

8. Closest option:

$$H_{\max} \approx 9.3 \text{ m.}$$

Source(s): Test MC exam_Corr / Test_Exam_all

O10. Question 10

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder is described by $U(t) = U_m \sin(\omega t)$ with $U_m = 0.9$ m/s, $T = 6.0$ s and $D = 0.60$ m. What is the Keulegan-Carpenter number?

Options:

- A. $KC = 6.0$
- B. $KC = 9.0$
- C. $KC = 12.0$
- D. $KC = 15.0$

Correct answer: B. $KC = 9.0$

Formulario / principles used

- Keulegan-Carpenter number:

$$KC = \frac{U_m T}{D}.$$

Step-by-step solution

1. Substitute the given values:

$$KC = \frac{(0.9)(6.0)}{0.60}.$$

2. Compute the numerator:

$$(0.9)(6.0) = 5.4.$$

3. Divide by the diameter:

$$KC = \frac{5.4}{0.60} = 9.0.$$

Source(s): 41107_mock_exam_with_answers

O11. Question 11

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. For $U_m = 0.8$ m/s, $D = 0.50$ m and $T = 4.0$ s, assume $C_D = 1.0$ and $C_M = 1.8$. What is the ratio $F_{I,\max}/F_{D,\max}$ approximately?

Options:

- A. 0.28
- B. 1.39
- C. 2.78
- D. 5.56

Correct answer: C. 2.78

Formulario / principles used

- Keulegan-Carpenter number:

$$KC = \frac{U_m T}{D}.$$

- Maximum inertia-to-drag ratio:

$$\frac{F_{I,\max}}{F_{D,\max}} = \frac{\pi^2}{KC} \frac{C_M}{C_D}.$$

Step-by-step solution

1. Compute KC :

$$KC = \frac{(0.8)(4.0)}{0.50} = \frac{3.2}{0.50} = 6.4.$$

2. Substitute into the ratio:

$$\frac{F_{I,\max}}{F_{D,\max}} = \frac{\pi^2}{6.4} \frac{1.8}{1.0}.$$

3. Compute each factor:

$$\frac{\pi^2}{6.4} = \frac{9.8696}{6.4} = 1.542, \quad \frac{1.8}{1.0} = 1.8.$$

4. Multiply:

$$\frac{F_{I,\max}}{F_{D,\max}} = (1.542)(1.8) = 2.78.$$

Source(s): 41107_mock_exam_with_answers

O12. Question 12

FLOW AROUND AND FORCES ON CYLINDERS. Plane oscillatory flow around a free smooth circular cylinder. The normalized fundamental lift frequency is $N_L = 2$. Which name is given to this flow regime?

Options:

- A. Single-pair regime
- B. Double-pair regime
- C. Three-pair regime
- D. No-separation regime

Correct answer: A. Single-pair regime **Table 4***

Formulario / principles used

- Normalized lift frequency:

$$N_L = \frac{f_L}{f_w}.$$

- Course convention:

$$N_L \approx 2 \Rightarrow \text{single-pair vortex shedding regime.}$$

Step-by-step solution

1. Given:

$$N_L = 2.$$

2. Match the value with the course regime map:

$$N_L = 2 \Rightarrow \text{single-pair regime.}$$

3. No further numerical calculation is required.

Source(s): 41107_mock_exam_with_answers

Froude-Krylov / diffraction / large-body questions

D1. Question 1

DIFFRACTION / LARGE BODIES. For wave diffraction around a cylinder, when is diffraction expected to be important?

Options:

- A. When $D/L \gtrsim 0.2$
- B. When $D/L \lesssim 0.02$
- C. Only when $KC > 20$
- D. Only when $Re < 1000$

Correct answer: A. When $D/L \gtrsim 0.2$

Formulario / principles used

- Diffraction parameter:

$$\frac{D}{L} = \frac{\text{characteristic body diameter}}{\text{wavelength}}.$$

- Course rule of thumb:

$$\frac{D}{L} \gtrsim O(0.2) \Rightarrow \text{diffraction is important.}$$

- If D/L is small, the body is slender relative to the wavelength and Morison/Froude-Krylov type loading is usually more appropriate.

Step-by-step solution

1. The question asks for the condition under which wave diffraction around a cylinder becomes important.
2. Diffraction becomes important when the cylinder is not small compared with the wavelength.
3. The course threshold is approximately:

$$\frac{D}{L} \gtrsim 0.2.$$

4. Option A states exactly this criterion.
5. Options B, C and D are traps: KC and Re are important for flow regime and viscous/separation effects, but the diffraction criterion is primarily based on D/L .

Source(s): 41107_mock_exam_with_answers / 41107_MCQ_Study_Guide

D2. Question 2

SHIP MOTION DYNAMICS - Froude-Krylov Force (Diffraction Neglected). For a ship in regular waves, under which assumption is the Froude-Krylov force dominant and diffraction can be neglected?

Options:

- A. $D/L > 0.5$ (large ship/short waves)

B. $D/L < 0.2$ (small body relative to wavelength)

C. $Fn > 0.3$ (high Froude number)

Correct answer: B. $D/L < 0.2$ (small body relative to wavelength)

Formulario / principles used

- Diffraction parameter:

$$\frac{D}{L} = \frac{\text{characteristic body dimension}}{\text{wavelength}}.$$

- Small-body / Morison-type regime:

$$\frac{D}{L} < 0.2 \Rightarrow \text{diffraction small.}$$

- Large-body / diffraction regime:

$$\frac{D}{L} \gtrsim 0.2 \Rightarrow \text{diffraction may be important.}$$

- Froude-Krylov force uses the undisturbed incident-wave pressure field.

Step-by-step solution

1. The Froude-Krylov force assumes the incident wave pressure field is not strongly modified by the body.
2. This assumption is most valid when the body is small compared with the wavelength.
3. The course rule of thumb is:

$$\frac{D}{L} < 0.2.$$

4. Therefore, diffraction can be neglected and Froude-Krylov force is dominant when:

$$D/L < 0.2.$$

5. Option A is the opposite case: a large body relative to the wavelength, where diffraction matters.
6. Option C uses Froude number, which is not the diffraction criterion in this context.

Source(s): 41107_MCQ_Study_Guide / Wave-structure interactions - Regular waves (L10)

Final formula reminder for this group

Steady current around cylinders

$$\begin{aligned}
 Re &= \frac{U_c D}{\nu} \\
 St &= \frac{f_v D}{U_c} \\
 f_{D,\text{osc}} &= 2f_v \\
 p_s - p_0 &= \frac{1}{2}\rho U_c^2 \\
 F_D &= \frac{1}{2}\rho C_D D U_c^2 \\
 C_D &= \frac{F_D}{\frac{1}{2}\rho U_c^2 D}
 \end{aligned}$$

Oscillatory flow and wave-related cylinder questions

$$\begin{aligned}
 U(t) &= U_m \sin(\omega t), \quad \omega = \frac{2\pi}{T} \\
 KC &= \frac{U_m T}{D} \\
 A &= \frac{\pi D^2}{4} \\
 F_{FK} &= \rho A \dot{U} \\
 F_{FK,\text{max}} &= \rho A U_m \omega \\
 F_x &= \frac{1}{2}\rho C_D D U |U| + \rho C_M A \dot{U} \\
 F_I &= \rho C_M A \dot{U}, \quad F_K = \rho A \dot{U}, \quad F_I = C_M F_K \\
 \frac{F_{I,\text{max}}}{F_{D,\text{max}}} &= \frac{\pi^2}{KC} \frac{C_M}{C_D} \\
 N_L &= \frac{f_L}{f_w} \\
 \omega^2 &= gk \tanh(kh), \quad L = \frac{2\pi}{k} \\
 H_{\text{max}} &\approx 0.142L \tanh(kh)
 \end{aligned}$$

Froude-Krylov and diffraction

$$\begin{aligned}
 F_{\text{exc}} &= F_{FK} + F_{\text{diff}} \\
 \frac{D}{L} < 0.2 &\Rightarrow \text{small body, diffraction usually negligible} \\
 \frac{D}{L} \gtrsim 0.2 &\Rightarrow \text{large-body diffraction may be important}
 \end{aligned}$$

41107 Marine and Ocean Engineering

Vortex-Induced Vibrations (VIV)

Solved MCQ Question Bank - professional formula notation

Scope and ordering

This PDF collects all reviewed questions beginning with **VORTEX INDUCED VIBRATIONS.** or **VIV.** Exact duplicates across the reviewed MCQ documents were removed. The questions are grouped by subtopic so the block is easier to study.

For each question, the structure is:

1. Original question statement and original answer options.
2. Correct answer highlighted immediately after the options.
3. **Formulario / principles used** with professional mathematical notation.
4. **Step-by-step solution** with numerical substitution whenever numerical data are present.

Default constants and notation

$$\rho = 1025 \text{ kg/m}^3, \quad \nu = 1.0 \times 10^{-6} \text{ m}^2/\text{s}, \quad A_c = \frac{\pi D^2}{4}$$

$$\omega = 2\pi f, \quad f = \frac{\omega}{2\pi}, \quad T = \frac{1}{f} = \frac{2\pi}{\omega}$$

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Natural frequency with added mass

N1. Question 1

VORTEX INDUCED VIBRATIONS. Calculate the un-damped natural frequency of a flexibly mounted circular cylinder in water where the following is given: the cylinder diameter $D = 0.2$ m, the mass (per unit length) $m = 4$ kg/m, the spring constant (per unit length) $k = 900$ kg/m and it is known that the normalized cross-flow vibration amplitude $A/D < 0.8$.

Options:

A. $f_n = 0.58$ Hz

B. $f_n = 2.39$ Hz

C. $f_n = 0.79$ Hz

Correct answer: C. $f_n = 0.79$ Hz

Note on the original stem. The original text writes $k = 900$ kg/m, but the calculation uses it as a stiffness per unit length, i.e. $k = 900$ N/m per metre span.

Formulario / principles used

- Cylinder cross-sectional area per unit length:

$$A_c = \frac{\pi D^2}{4}$$

- Added mass per unit length:

$$m' = \rho C_m A_c$$

For a circular cylinder in the potential-flow / small-amplitude estimate, use $C_m \approx 1$.

- Effective mass per unit length:

$$m_{\text{tot}} = m + m'$$

- Undamped angular natural frequency:

$$\omega_n = \sqrt{\frac{k}{m_{\text{tot}}}}$$

- Natural frequency in hertz:

$$f_n = \frac{\omega_n}{2\pi}$$

Step-by-step solution

1. Compute the cylinder area:

$$A_c = \frac{\pi D^2}{4} = \frac{\pi(0.2)^2}{4} = 0.0314 \text{ m}^2.$$

2. Compute the added mass per unit length:

$$m' = \rho C_m A_c = (1025)(1)(0.0314) = 32.2 \text{ kg/m}.$$

3. Compute the total effective mass:

$$m_{\text{tot}} = m + m' = 4.0 + 32.2 = 36.2 \text{ kg/m}.$$

4. Compute the angular natural frequency:

$$\omega_n = \sqrt{\frac{k}{m_{\text{tot}}}} = \sqrt{\frac{900}{36.2}} = \sqrt{24.86} = 4.99 \text{ rad/s}.$$

5. Convert from angular frequency to hertz:

$$f_n = \frac{\omega_n}{2\pi} = \frac{4.99}{2\pi} = 0.794 \text{ Hz} \approx 0.79 \text{ Hz}.$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test_Exam_all / 41107_MCQ_Study_Guide

N2. Question 2

VORTEX INDUCED VIBRATIONS. Calculate the undamped natural frequency of a flexibly mounted circular cylinder in water. Take $D = 0.08 \text{ m}$, structural mass per unit length $m = 4.0 \text{ kg/m}$, spring constant per unit length $k = 900 \text{ N/m}$, and assume $A/D < 0.8$ so that $C_m = 1$ for the added mass estimate. What is f_n approximately?

Options:

- A. 0.79 Hz
- B. 1.58 Hz
- C. 2.25 Hz
- D. 3.16 Hz

Correct answer: B. 1.58 Hz

Formulario / principles used

$$A_c = \frac{\pi D^2}{4}, \quad m' = \rho C_m A_c, \quad m_{\text{tot}} = m + m'$$

$$\omega_n = \sqrt{\frac{k}{m_{\text{tot}}}}, \quad f_n = \frac{\omega_n}{2\pi}$$

Step-by-step solution

1. Compute the cross-sectional area:

$$A_c = \frac{\pi(0.08)^2}{4} = 0.00503 \text{ m}^2.$$

2. Compute added mass per unit length:

$$m' = \rho C_m A_c = (1025)(1)(0.00503) = 5.15 \text{ kg/m}.$$

3. Compute total effective mass:

$$m_{\text{tot}} = m + m' = 4.0 + 5.15 = 9.15 \text{ kg/m}.$$

4. Compute the angular natural frequency:

$$\omega_n = \sqrt{\frac{900}{9.15}} = \sqrt{98.36} = 9.92 \text{ rad/s}.$$

5. Convert to hertz:

$$f_n = \frac{9.92}{2\pi} = 1.58 \text{ Hz}.$$

Source(s): 41107_mock_exam_with_answers

Free decay damping

D1. Question 3

VORTEX INDUCED VIBRATIONS. The amplitude response as function of time in a free decay test in water is shown in the following figure. What is the total damping factor (ζ) and the un-damped natural angular frequency (ω_n) when $y_n = 2$, $y_{n+1} = 0.5$ and $T = 1$ s?

Options:

- A. $\zeta = 0.096$; $\omega_n = 6.25$ rad/s
- B. $\zeta = 0.318$; $\omega_n = 5.96$ rad/s
- C. $\zeta = 0.221$; $\omega_n = 6.13$ rad/s

Correct answer: C. $\zeta = 0.221$; $\omega_n = 6.13$ rad/s

Formulario / principles used

- Logarithmic decrement from two successive peaks:

$$\delta = \ln \left(\frac{y_n}{y_{n+1}} \right)$$

- Course shortcut for damping factor:

$$\zeta \approx \frac{\delta}{2\pi}$$

- Damped angular frequency from the measured period:

$$\omega_d = \frac{2\pi}{T}$$

- Standard exact relation:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \quad \Rightarrow \quad \omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}}$$

Step-by-step solution

1. Compute the logarithmic decrement:

$$\delta = \ln \left(\frac{y_n}{y_{n+1}} \right) = \ln \left(\frac{2}{0.5} \right) = \ln(4) = 1.386.$$

2. Compute the damping factor using the course shortcut:

$$\zeta \approx \frac{\delta}{2\pi} = \frac{1.386}{2\pi} = 0.221.$$

3. Compute the damped angular frequency from the period:

$$\omega_d = \frac{2\pi}{T} = \frac{2\pi}{1} = 6.28 \text{ rad/s.}$$

4. Compare with the answer options. Only option C contains the correct damping factor:

$$\zeta = 0.221.$$

5. Therefore the intended MCQ answer is option C.

Important note. With the strict standard vibration relation, $\omega_n = \omega_d / \sqrt{1 - \zeta^2}$, the numerical value would be $6.28 / \sqrt{1 - 0.221^2} = 6.44$ rad/s. Therefore, the frequency value in the original option appears to follow the answer-set/course convention rather than the strict exact relation. For the MCQ, select option C.

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test_Exam_all / 41107_MCQ_Study_Guide

D2. Question 4

VORTEX INDUCED VIBRATIONS. The amplitude response in a free-decay test is shown in the figure. If $y_n = 1.5$, $y_{n+1} = 0.9$ and $T = 0.8$ s, what are the damping ratio ζ and the undamped natural angular frequency ω_n approximately?

Options:

- A. $\zeta = 0.081$, $\omega_n = 7.88$ rad/s
- B. $\zeta = 0.081$, $\omega_n = 6.28$ rad/s
- C. $\zeta = 0.163$, $\omega_n = 7.88$ rad/s
- D. $\zeta = 0.163$, $\omega_n = 6.28$ rad/s

Correct answer: A. $\zeta = 0.081$, $\omega_n = 7.88$ rad/s

Formulario / principles used

$$\delta = \ln \left(\frac{y_n}{y_{n+1}} \right), \quad \zeta \approx \frac{\delta}{2\pi}$$

$$\omega_d = \frac{2\pi}{T}, \quad \omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}}$$

Step-by-step solution

1. Compute logarithmic decrement:

$$\delta = \ln \left(\frac{1.5}{0.9} \right) = \ln(1.6667) = 0.5108.$$

2. Estimate damping ratio:

$$\zeta \approx \frac{0.5108}{2\pi} = 0.0813 \approx 0.081.$$

3. Compute damped angular frequency:

$$\omega_d = \frac{2\pi}{0.8} = 7.854 \text{ rad/s.}$$

4. Correct to undamped angular natural frequency:

$$\omega_n = \frac{7.854}{\sqrt{1 - 0.0813^2}} = 7.88 \text{ rad/s.}$$

5. Therefore the correct option is A.

Source(s): 41107_mock_exam_with_answers

Reduced velocity

R1. Question 5

VORTEX INDUCED VIBRATIONS. Plane oscillatory flow around a free, smooth, flexibly-mounted, circular cylinder. What is the reduced velocity when the oscillatory flow is defined by $U[\text{m/s}] = 2.5 \cdot \sin(\omega t)$, $D = 0.5 \text{ m}$, and the angular natural frequency $\omega_n = 7.85 \text{ rad/s}$?

Options:

- A. $V_r = 6.25$
- B. $V_r = 0.25$
- C. $V_r = 4.0$

Correct answer: C. $V_r = 4.0$

Formulario / principles used

- Natural frequency in hertz:

$$f_n = \frac{\omega_n}{2\pi}$$

- Reduced velocity in oscillatory flow:

$$V_r = \frac{U_m}{Df_n}$$

where U_m is the maximum oscillatory velocity, not the mean velocity.

Step-by-step solution

- Read the velocity amplitude from $U(t)$:

$$U(t) = 2.5 \sin(\omega t) \Rightarrow U_m = 2.5 \text{ m/s.}$$

- Convert angular natural frequency to ordinary frequency:

$$f_n = \frac{\omega_n}{2\pi} = \frac{7.85}{2\pi} = 1.25 \text{ Hz.}$$

- Compute the natural-frequency velocity scale:

$$Df_n = (0.5)(1.25) = 0.625 \text{ m/s.}$$

- Compute reduced velocity:

$$V_r = \frac{U_m}{Df_n} = \frac{2.5}{0.625} = 4.0.$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test_Exam_all / 41107_MCQ_Study_Guide

R2. Question 6

VORTEX INDUCED VIBRATIONS. Plane oscillatory flow around a free, smooth, flexibly-mounted, circular cylinder. What is the reduced velocity when the oscillatory flow is defined by $U[\text{m/s}] = 0.5 \cdot \sin(\omega t)$, $D = 1 \text{ m}$, and the angular natural frequency $\omega_n = 15.71 \text{ rad/s}$?

Options:

- A. $V_r = 0.8$

B. $V_r = 1.25$

C. $V_r = 0.2$

Correct answer: C. $V_r = 0.2$

Formulario / principles used

$$f_n = \frac{\omega_n}{2\pi}, \quad V_r = \frac{U_m}{Df_n}$$

Step-by-step solution

1. Extract the maximum velocity:

$$U(t) = 0.5 \sin(\omega t) \Rightarrow U_m = 0.5 \text{ m/s.}$$

2. Convert angular natural frequency:

$$f_n = \frac{15.71}{2\pi} = 2.50 \text{ Hz.}$$

3. Compute the velocity scale:

$$Df_n = (1)(2.50) = 2.50 \text{ m/s.}$$

4. Compute the reduced velocity:

$$V_r = \frac{0.5}{2.50} = 0.20.$$

Source(s): Test MC exam_Corr / Test_Exam_all

R3. Question 7

VORTEX INDUCED VIBRATIONS. Plane oscillatory flow around a free, smooth, flexibly-mounted circular cylinder. The oscillatory flow is $U(t) = 1.8 \sin(\omega t)$ m/s, the diameter is $D = 0.40$ m, and the angular natural frequency is $\omega_n = 9.42$ rad/s. What is the reduced velocity V_r ?

Options:

A. $V_r = 2.0$

B. $V_r = 3.0$

C. $V_r = 4.5$

D. $V_r = 6.0$

Correct answer: B. $V_r = 3.0$

Formulario / principles used

$$f_n = \frac{\omega_n}{2\pi}, \quad V_r = \frac{U_m}{Df_n}$$

Step-by-step solution

1. Extract the maximum velocity:

$$U(t) = 1.8 \sin(\omega t) \quad \Rightarrow \quad U_m = 1.8 \text{ m/s.}$$

2. Convert angular frequency to hertz:

$$f_n = \frac{9.42}{2\pi} = 1.50 \text{ Hz.}$$

3. Compute Df_n :

$$Df_n = (0.40)(1.50) = 0.60 \text{ m/s.}$$

4. Compute reduced velocity:

$$V_r = \frac{1.8}{0.60} = 3.0.$$

Source(s): 41107_mock_exam_with_answers

Amplitude ratios and vibration direction

A1. Question 8

VORTEX INDUCED VIBRATIONS. Steady current around a free, flexibly-mounted circular cylinder. What is the ratio between the maximum in-line vibration amplitude and the maximum cross-flow vibration amplitude, A_x/A_y ? (values below 1 indicate cross-flow vibration amplitude is larger than in-line vibration amplitude)?

Options:

- A. Approximately 1/10
- B. Approximately 10
- C. Approximately 1

Correct answer: A. Approximately 1/10

Formulario / principles used

- Typical Vortex-Induced Vibrations (VIV) amplitude relation:

$$\frac{A_x}{A_y} \approx 0.1 = \frac{1}{10}.$$

- A_x : in-line vibration amplitude, mainly related to oscillating drag.
- A_y : cross-flow vibration amplitude, mainly related to oscillating lift.

Step-by-step solution

1. Identify the requested ratio:

$$\frac{A_x}{A_y} = \frac{\text{in-line amplitude}}{\text{cross-flow amplitude}}.$$

2. In typical VIV, cross-flow vibration is much larger than in-line vibration:

$$A_y \approx 10A_x.$$

3. Rearrange to the requested ratio:

$$\frac{A_x}{A_y} \approx \frac{A_x}{10A_x} = \frac{1}{10} = 0.1.$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test_Exam_all / 41107_MCQ_Study_Guide

A2. Question 9

VORTEX INDUCED VIBRATIONS. Steady current around a free, flexibly-mounted circular cylinder. What is the ratio between the maximum in-line vibration amplitude and the maximum cross-flow vibration amplitude, A_x/A_y ? (values below 1 indicate cross-flow vibration amplitude is larger than in-line vibration amplitude)?

Options:

- A. approximately 1/10
- B. approximately 1

C. approximately 1/100

Correct answer: A. approximately 1/10

Formulario / principles used

$$\frac{A_x}{A_y} \approx 0.1 = \frac{1}{10}$$

Values below 1 mean that $A_y > A_x$, i.e. cross-flow amplitude is larger.

Step-by-step solution

1. The question asks for A_x/A_y , not A_y/A_x .

2. Typical VIV gives:

$$A_x \approx 0.1A_y.$$

3. Substitute directly into the ratio:

$$\frac{A_x}{A_y} \approx \frac{0.1A_y}{A_y} = 0.1 = \frac{1}{10}.$$

4. Therefore option A is correct.

Source(s): Test MC exam_Corr / Test_Exam_all

Stability parameter and amplitude response

S1. Question 10

VORTEX INDUCED VIBRATIONS. Steady current around a free, flexibly-mounted, smooth, circular cylinder. The mass ratio $M = 5$, and the structural damping factor $\zeta_s = 0.05$. What is the expected cross-flow amplitude response assuming the following model for the amplitude response as function of stability parameter: $A/D = 1.5/K_s$?

Options:

A. $A/D = 0.72$

B. $A/D = 0.96$

C. $A/D = 0.48$

Correct answer: C. $A/D = 0.48$

Formulario / principles used

- Structural logarithmic decrement:

$$\delta_s = 2\pi\zeta_s$$

- Stability parameter:

$$K_s = 2M\delta_s = 4\pi M\zeta_s$$

- Given empirical amplitude model:

$$\frac{A}{D} = \frac{1.5}{K_s}$$

Step-by-step solution

1. Compute the logarithmic decrement:

$$\delta_s = 2\pi\zeta_s = 2\pi(0.05) = 0.314.$$

2. Compute the stability parameter:

$$K_s = 2M\delta_s = 2(5)(0.314) = 3.14.$$

3. Use the amplitude model from the question:

$$\frac{A}{D} = \frac{1.5}{K_s} = \frac{1.5}{3.14} = 0.477.$$

4. Round to the closest option:

$$\frac{A}{D} \approx 0.48.$$

Source(s): Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

S2. Question 11

VIV - Stability Parameter and Suppression. A marine riser with the following properties: $D = 0.3$ m, structural mass per unit length $m = 50$ kg/m, structural damping $\zeta_s = 0.01$. Will VIV be a concern?

Options:

- A.** No - the stability parameter is large enough to suppress VIV
- B.** Yes - the stability parameter is small, significant VIV is expected
- C.** Cannot determine without current velocity

Correct answer: B. Yes - the stability parameter is small, significant VIV is expected

Formulario / principles used

- Displaced-fluid mass per unit length using the source's simplified convention:

$$m_d = \rho A_c = \rho \frac{\pi D^2}{4}$$

- Mass ratio:

$$M = \frac{m}{m_d}$$

- Structural logarithmic decrement and stability parameter:

$$\delta_s = 2\pi\zeta_s, \quad K_s = 2M\delta_s$$

- Interpretation: small K_s means low damping/light structure and higher VIV susceptibility.

Step-by-step solution

1. Compute the displaced-fluid mass per unit length:

$$m_d = \rho \frac{\pi D^2}{4} = 1025 \frac{\pi (0.3)^2}{4} = 72.4 \text{ kg/m.}$$

2. Compute the mass ratio:

$$M = \frac{m}{m_d} = \frac{50}{72.4} = 0.69.$$

3. Compute structural logarithmic decrement:

$$\delta_s = 2\pi\zeta_s = 2\pi(0.01) = 0.0628.$$

4. Compute stability parameter:

$$K_s = 2M\delta_s = 2(0.69)(0.0628) = 0.087.$$

5. Since K_s is very small, the riser is not strongly damped or heavy enough to suppress VIV.

6. Therefore significant VIV is expected.

Source(s): 41107_MCQ_Study_Guide

In-line instability regions and lock-in

I1. Question 12

VORTEX INDUCED VIBRATIONS. Steady current around a free, flexibly-mounted circular cylinder. Which in-line instability region (vibration) is associated with normal vortex shedding when $St = 0.2$?

Options:

- A. 2nd instability region ($V_r \approx 2.5$)
- B. 3rd instability region ($V_r \approx 5$)
- C. 1st instability region ($V_r \approx 1.7$)

Correct answer: C. 1st instability region ($V_r \approx 1.7$)

Formulario / principles used

- Strouhal relation:

$$St = \frac{f_v D}{U}$$

- Reduced velocity:

$$V_r = \frac{U}{D f_n}$$

- Cross-flow lock-in scale:

$$V_r \approx \frac{1}{St}$$

For $St = 0.2$, this gives $V_r \approx 5$, which is the cross-flow lock-in scale, not the requested in-line instability region.

- Course MCQ convention: the in-line instability associated with normal vortex shedding is the 1st in-line instability region around $V_r \approx 1.7$.

Step-by-step solution

1. The question specifically asks for an **in-line** instability region, not the cross-flow lock-in region.
2. Check the common cross-flow lock-in scale:

$$V_r \approx \frac{1}{St} = \frac{1}{0.2} = 5.$$

3. This value corresponds to the cross-flow lock-in scale, so option B is the typical trap.
4. In the course convention, normal vortex shedding is associated with the 1st in-line instability region.
5. Therefore the correct option is:

1st instability region ($V_r \approx 1.7$).

Source(s): Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

Final formula reminder for this group

Natural frequency with added mass

$$A_c = \frac{\pi D^2}{4}$$

$$m' = \rho C_m A_c$$

$$m_{\text{tot}} = m + m'$$

$$\omega_n = \sqrt{\frac{k}{m_{\text{tot}}}}, \quad f_n = \frac{\omega_n}{2\pi}$$

Free decay damping

$$\delta = \ln \left(\frac{y_n}{y_{n+1}} \right)$$

$$\zeta \approx \frac{\delta}{2\pi}$$

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} \quad (\text{exact logarithmic decrement relation})$$

$$\omega_d = \frac{2\pi}{T_d}, \quad \omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}}$$

Reduced velocity and lock-in

$$f_n = \frac{\omega_n}{2\pi}$$

$$V_r = \frac{U}{Df_n} \quad (\text{steady current}), \quad V_r = \frac{U_m}{Df_n} \quad (\text{oscillatory flow})$$

$$St = \frac{f_v D}{U}$$

$$\text{Cross-flow lock-in scale: } V_r \approx \frac{1}{St}$$

Amplitude response and stability parameter

$$\delta_s = 2\pi\zeta_s$$

$$K_s = 2M\delta_s = 4\pi M\zeta_s$$

$$\frac{A}{D} = \frac{1.5}{K_s} \quad \text{when this empirical model is given}$$

$$\frac{A_x}{A_y} \approx 0.1 = \frac{1}{10}$$

41107 Marine and Ocean Engineering

Waves / Wave Theory / Wave Statistics / Spectral Analysis

Solved MCQ Question Bank - professional formula notation

Scope and ordering

This PDF collects the reviewed questions belonging to the **Waves** block, including **wave theory**, **wave statistics**, **spectral moments** and **spectral analysis**. Exact duplicates across the reviewed MCQ documents were removed. Same pattern with different numerical values is kept as a separate variant. The questions are ordered as follows:

1. Wave theory and regular-wave kinematics.
2. Wave statistics and extreme values.
3. Spectral moments and spectral response.

For each question, the structure is:

1. Original question statement and original answer options.
2. Correct answer highlighted immediately after the options.
3. **Formulario / principles used** with professional mathematical notation.
4. **Step-by-step solution** with numerical substitution whenever numerical data are present.

Default constants and notation

$$g = 9.81 \text{ m/s}^2, \quad \rho = 1025 \text{ kg/m}^3, \quad 1 \text{ knot} = 0.51444 \text{ m/s}$$

$$\omega = \frac{2\pi}{T} = 2\pi f, \quad f = \frac{1}{T} = \frac{\omega}{2\pi}, \quad k = \frac{2\pi}{L}$$

$$\text{Deep water: } L_0 = \frac{gT^2}{2\pi}, \quad c = \frac{L}{T} = \frac{\omega}{k}$$

Contents

1 Wave theory and regular-wave kinematics	2
2 Wave statistics and extreme values	8
3 Spectral moments and spectral response	13
4 Final formula reminder for this group	19

Wave theory and regular-wave kinematics

W1. Deep versus shallow water

WAVE THEORY - Deep vs Shallow Water. For deep-water waves, which of the following best describes the relationship between wavelength L and water depth h ?

Options:

- A. $h > L/20$ (deep water: $\tanh(kh) \approx 1$)
- B. $h < L/20$ (shallow water: $\tanh(kh) \approx kh$)
- C. $h = L/2$ (intermediate depth)

Correct answer: A. $h > L/20$ (as labelled in the source question)

Important correction. The physically standard criterion is $h > L/2$ for deep water and $h < L/20$ for shallow water. The source option A is internally inconsistent because it writes $h > L/20$ but labels it as deep water with $\tanh(kh) \approx 1$. For exam solving, remember the physical rule: $h > L/2 \Rightarrow$ deep water.

Formulario / principles used

- General dispersion relation:

$$\omega^2 = gk \tanh(kh)$$

- Wave number:

$$k = \frac{2\pi}{L}$$

- Deep-water limit:

$$h > \frac{L}{2} \Rightarrow kh > \pi, \quad \tanh(kh) \approx 1$$

- Shallow-water limit:

$$h < \frac{L}{20} \Rightarrow kh < \frac{\pi}{10}, \quad \tanh(kh) \approx kh$$

Step-by-step solution

1. Deep water means that the seabed has almost no influence on wave motion.
2. Mathematically, that corresponds to:

$$\tanh(kh) \approx 1.$$

3. Since $k = 2\pi/L$, deep-water behaviour requires:

$$kh = \frac{2\pi h}{L} \gtrsim \pi.$$

4. Therefore:

$$\frac{2\pi h}{L} \gtrsim \pi \Rightarrow h \gtrsim \frac{L}{2}.$$

5. Among the given options, the source labels option A as the intended deep-water option, but the correct physical criterion is $h > L/2$.

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

W2. Deep-water phase speed and group velocity

WAVE THEORY - Deep Water Wave Speed and Group Velocity. A regular deep-water wave has period $T = 10$ s. What is the phase speed c and the group velocity c_g ?

Options:

- A. $c = 15.6$ m/s, $c_g = 7.8$ m/s
- B. $c = 7.8$ m/s, $c_g = 15.6$ m/s
- C. $c = 15.6$ m/s, $c_g = 15.6$ m/s

Correct answer: A. $c = 15.6$ m/s, $c_g = 7.8$ m/s

Formulario / principles used

- Deep-water wavelength:

$$L_0 = \frac{gT^2}{2\pi}$$

- Phase speed:

$$c = \frac{L_0}{T} = \frac{gT}{2\pi}$$

- Deep-water group velocity:

$$c_g = \frac{c}{2}$$

Step-by-step solution

1. Calculate phase speed:

$$c = \frac{gT}{2\pi} = \frac{9.81 \cdot 10}{2\pi} = \frac{98.1}{6.283} = 15.6 \text{ m/s.}$$

2. For deep-water waves, the group velocity is half the phase speed:

$$c_g = \frac{c}{2} = \frac{15.6}{2} = 7.8 \text{ m/s.}$$

3. Therefore the correct pair is:

$$c = 15.6 \text{ m/s,} \quad c_g = 7.8 \text{ m/s.}$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

W3. Maximum horizontal particle velocity at the surface

WAVE THEORY - Particle Velocity and Acceleration in Regular Waves. A regular deep-water wave has height $H = 2$ m, period $T = 8$ s. What is the maximum horizontal particle velocity U_m at the surface ($z = 0$)?

Options:

- A. $U_m = 0.39$ m/s
- B. $U_m = 0.785$ m/s
- C. $U_m = 1.57$ m/s

Correct answer: B. $U_m = 0.785$ m/s

Formulario / principles used

- Wave amplitude:

$$a = \frac{H}{2}$$

- Angular frequency:

$$\omega = \frac{2\pi}{T}$$

- Deep-water horizontal velocity amplitude at the free surface:

$$U_m = \omega a = \frac{2\pi}{T} \frac{H}{2}$$

Step-by-step solution

- Convert wave height to wave amplitude:

$$a = \frac{H}{2} = \frac{2}{2} = 1.0 \text{ m.}$$

- Compute angular frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{8} = 0.785 \text{ rad/s.}$$

- Compute the maximum horizontal particle velocity at the surface:

$$U_m = \omega a = (0.785)(1.0) = 0.785 \text{ m/s.}$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

W4. Maximum wave height before breaking

FLOW AROUND AND FORCES ON CYLINDERS. What is the maximum wave height on a horizontal bed given a wave period $T = 7$ s and a water depth $h = 19$ m?

Options:

- A. $H_{\max} = 15.2$ m
- B. $H_{\max} = 9.3$ m
- C. $H_{\max} = 4.1$ m

Correct answer: B. $H_{\max} = 9.3$ m

Formulario / principles used

- Angular frequency:

$$\omega = \frac{2\pi}{T}$$

- Linear wave dispersion relation:

$$\omega^2 = gk \tanh(kh)$$

- Wavelength:

$$L = \frac{2\pi}{k}$$

- Miche breaking criterion:

$$H_{\max} = 0.142 L \tanh(kh)$$

Step-by-step solution

1. Calculate angular frequency:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{7} = 0.898 \text{ rad/s.}$$

2. Solve the dispersion relation numerically:

$$\omega^2 = gk \tanh(kh)$$

$$(0.898)^2 = 9.81 k \tanh(19k).$$

3. This gives approximately:

$$k \approx 0.0847 \text{ m}^{-1}.$$

4. Calculate the wavelength:

$$L = \frac{2\pi}{k} = \frac{2\pi}{0.0847} = 74.2 \text{ m.}$$

5. Calculate
- kh
- and
- $\tanh(kh)$
- :

$$kh = (0.0847)(19) = 1.61,$$

$$\tanh(kh) = \tanh(1.61) \approx 0.92.$$

6. Apply Miche criterion:

$$H_{\max} = 0.142(74.2)(0.92) = 9.69 \text{ m.}$$

7. The closest available answer is:

$$H_{\max} = 9.3 \text{ m.}$$

Source(s): *Test_Exam_all / Block 2 reference PDF*

W5. True wave period from encountered period

WAVES: A ship sails with a speed of 18 knots. The ship encounters regular waves at an angle $\beta = 135^\circ$ (bow-quartering waves). On the ship, the encountered wave period is $T_{enc} = 10$ s. What is the absolute (true) wave period T , as measured by a fixed observer relative to the waves?

Options:

A. $T = 12.5$ s

B. $T = 13.0$ s

C. $T = 13.5$ s

Correct answer: B. $T = 13.0$ s

Formulario / principles used

- Encounter frequency in deep water:

$$\omega_e = \omega - \frac{\omega^2}{g} U \cos \beta$$

- Encounter period:

$$\omega_e = \frac{2\pi}{T_{enc}}$$

- True wave period:

$$T = \frac{2\pi}{\omega}$$

- Speed conversion:

$$U \text{ [m/s]} = 0.51444 U \text{ [knots]}$$

Step-by-step solution

1. Convert the ship speed:

$$U = 18(0.51444) = 9.26 \text{ m/s.}$$

2. Convert encountered period to encountered angular frequency:

$$\omega_e = \frac{2\pi}{T_{enc}} = \frac{2\pi}{10} = 0.628 \text{ rad/s.}$$

3. Compute the heading cosine:

$$\cos(135^\circ) = -0.707.$$

4. Insert into the encounter-frequency relation:

$$0.628 = \omega - \frac{\omega^2}{9.81}(9.26)(-0.707).$$

5. Since the cosine is negative, the second term becomes positive:

$$0.628 = \omega + 0.667\omega^2.$$

6. Rearrange as a quadratic:

$$0.667\omega^2 + \omega - 0.628 = 0.$$

7. Solve for the positive root:

$$\omega = \frac{-1 + \sqrt{1 + 4(0.667)(0.628)}}{2(0.667)} = 0.477 \text{ rad/s.}$$

8. Convert to true period:

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{0.477} = 13.18 \text{ s.}$$

9. The closest option is:

$$T = 13.0 \text{ s.}$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

W6. Encountered period in head seas

WAVE THEORY - Encountered Period (Head Seas). A ship travels at 20 knots in head seas ($\beta = 180^\circ$). The waves have frequency $f = 0.1$ Hz. What is the encountered period T_e ?

Options:

A. $T_e = 8.2$ s

B. $T_e = 6.9$ s

C. $T_e = 6.0$ s

Correct answer: C. $T_e = 6.0$ s

Formulario / principles used

- Wave angular frequency:

$$\omega = 2\pi f$$

- Encounter angular frequency:

$$\omega_e = \omega - \frac{\omega^2}{g} U \cos \beta$$

- Head seas:

$$\beta = 180^\circ, \quad \cos(180^\circ) = -1$$

- Encountered period:

$$T_e = \frac{2\pi}{\omega_e}$$

Step-by-step solution

- Convert the ship speed:

$$U = 20(0.51444) = 10.29 \text{ m/s.}$$

- Compute wave angular frequency:

$$\omega = 2\pi f = 2\pi(0.1) = 0.628 \text{ rad/s.}$$

- In head seas, $\cos \beta = -1$, so:

$$\omega_e = \omega + \frac{\omega^2}{g} U.$$

- Substitute numerically:

$$\omega_e = 0.628 + \frac{(0.628)^2}{9.81}(10.29).$$

- Compute the added encounter-frequency term:

$$\frac{(0.628)^2}{9.81}(10.29) = \frac{0.394}{9.81}(10.29) = 0.414 \text{ rad/s.}$$

- Therefore:

$$\omega_e = 0.628 + 0.414 = 1.042 \text{ rad/s.}$$

- Convert to encountered period:

$$T_e = \frac{2\pi}{1.042} = 6.03 \text{ s} \approx 6.0 \text{ s.}$$

Wave statistics and extreme values

S1. Significant wave height from a wave record

WAVES: From a wave record consisting of 141 waves (trough to crest), the number of waves of various heights are as reported in the table below. What is the significant wave height?

H [m]	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
N	11	38	45	27	12	4	3	1

Options:

A. $H_s = 2.25$ m

B. $H_s = 2.15$ m

C. $H_s = 2.35$ m

Correct answer: C. $H_s = 2.35$ m

Formulario / principles used

- Significant wave height from a wave record:

$$H_s = H_{1/3} = \text{mean height of the highest one-third of all waves}$$

- Number of waves in the top third:

$$N_{1/3} = \frac{N_{\text{total}}}{3}$$

Step-by-step solution

- Compute the number of waves in the highest one-third:

$$N_{1/3} = \frac{141}{3} = 47 \text{ waves.}$$

- Select the 47 highest waves, starting from the largest wave-height class:

$$4.0 \text{ m} : 1, \quad 3.5 \text{ m} : 3, \quad 3.0 \text{ m} : 4, \quad 2.5 \text{ m} : 12.$$

- Count these waves:

$$1 + 3 + 4 + 12 = 20 \text{ waves.}$$

- More waves needed:

$$47 - 20 = 27 \text{ waves.}$$

- The next class is 2.0 m, which contains exactly 27 waves.

- Compute the mean height of the selected 47 waves:

$$H_s = \frac{1(4.0) + 3(3.5) + 4(3.0) + 12(2.5) + 27(2.0)}{47}.$$

- Substitute and add:

$$H_s = \frac{4.0 + 10.5 + 12.0 + 30.0 + 54.0}{47} = \frac{110.5}{47} = 2.35 \text{ m.}$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

S2. Significant wave height from a sketched spectrum

WAVE STATISTICS - Significant Wave Height from Spectrum. A wave spectrum $S(\omega)$ has been sketched. The peak spectral density is $S_{\max} \approx 5 \text{ m}^2\text{s}/\text{rad}$ and the area under the curve is approximately $m_0 \approx 1 \text{ m}^2$. Estimate H_s .

Options:

A. $H_s = 2 \text{ m}$

B. $H_s = 6 \text{ m}$

C. $H_s = 4 \text{ m}$

Correct answer: C. $H_s = 4 \text{ m}$

Formulario / principles used

- Zeroth spectral moment:

$$m_0 = \int_0^\infty S(\omega) d\omega$$

- Significant wave height from spectral moment:

$$H_s = 4\sqrt{m_0}$$

Step-by-step solution

1. Interpret the area under the wave spectrum as the zeroth spectral moment:

$$m_0 = \int_0^\infty S(\omega) d\omega.$$

2. The question states that the area is approximately:

$$m_0 \approx 1 \text{ m}^2.$$

3. Use the significant-wave-height relation:

$$H_s = 4\sqrt{m_0}.$$

4. Substitute numerically:

$$H_s = 4\sqrt{1} = 4 \text{ m}.$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

S3. Crest exceedance from Rayleigh distribution

WAVE STATISTICS. For an irregular wave record, the variance of the surface elevation is $m_0 = 1.44 \text{ m}^2$. What is the probability that a wave crest exceeds the level $h_1 = 1.2 \text{ m}$?

Options:

A. 13.5%

B. 36.8%

C. 60.7%

D. 77.9%

Correct answer: C. 60.7%

Formulario / principles used For Rayleigh-distributed crest amplitudes:

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2m_0}\right)$$

where m_0 is the variance of the surface elevation and h_1 is the crest level threshold.

Step-by-step solution

1. Square the threshold level:

$$h_1^2 = (1.2)^2 = 1.44 \text{ m}^2.$$

2. Compute the denominator:

$$2m_0 = 2(1.44) = 2.88 \text{ m}^2.$$

3. Substitute in the Rayleigh exceedance formula:

$$P(Z > h_1) = \exp\left(-\frac{1.44}{2.88}\right) = \exp(-0.5).$$

4. Compute the probability:

$$P(Z > h_1) = 0.607 = 60.7\%.$$

Source(s): 41107_mock_exam_with_answers / Block 2 reference PDF

S4. Gaussian exceedance from acceleration variance

WAVE STATISTICS - Rayleigh Distribution & Exceedance Probability. An acceleration spectrum has variance $\sigma^2 = m_0 = 1.5 \text{ (m/s}^2\text{)}^2$. What is the probability of exceeding an acceleration level $a = 2.5 \text{ m/s}^2$?

Options:

A. $p_a = 5\%$

B. $p_a = 12\%$

C. $p_a \approx 2\%$

Correct answer: C. $p_a \approx 2\%$

Important distinction. The source question labels Rayleigh, but the selected answer follows a Gaussian process-value exceedance using a standard normal table. A Rayleigh peak exceedance with the same numbers would give approximately 12.5%.

Formulario / principles used

- Standard deviation from variance:

$$\sigma = \sqrt{m_0}$$

- Standard normal variable:

$$z = \frac{a}{\sigma}$$

- Gaussian exceedance probability:

$$P(A > a) = P(Z > z)$$

Step-by-step solution

1. Compute the standard deviation:

$$\sigma = \sqrt{m_0} = \sqrt{1.5} = 1.225 \text{ m/s}^2.$$

2. Compute the standard normal value:

$$z = \frac{a}{\sigma} = \frac{2.5}{1.225} = 2.04.$$

3. From a standard normal table:

$$P(Z > 2.04) \approx 0.021.$$

4. Convert to percent:

$$0.021 = 2.1\% \approx 2\%.$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

S5. Most probable largest wave height

SPECTRAL ANALYSIS - Most Probable Largest Wave Height. A 3-hour storm sea state has $H_s = 4$ m and mean zero-crossing period $T_z = 8$ s. What is the most probable largest wave height $H_{\max}$ in this storm?

Options:

A. $H_{\max} \approx 6.3$ m

B. $H_{\max} \approx 7.4$ m

C. $H_{\max} \approx 8.5$ m

Correct answer: B. $H_{\max} \approx 7.4$ m

Formulario / principles used

- Number of waves during the storm:

$$N = \frac{T_{\text{storm}}}{T_z}$$

- Most probable largest wave height:

$$H_{\text{MPL}} = \frac{H_s}{2} \sqrt{2 \ln N}$$

Step-by-step solution

1. Convert storm duration to seconds:

$$T_{\text{storm}} = 3 \text{ h} = 3(3600) = 10800 \text{ s.}$$

2. Compute the number of waves:

$$N = \frac{10800}{8} = 1350.$$

3. Compute the logarithmic factor:

$$\ln N = \ln(1350) = 7.21.$$

4. Compute the square-root term:

$$\sqrt{2 \ln N} = \sqrt{2(7.21)} = \sqrt{14.42} = 3.80.$$

5. Compute the most probable largest wave height:

$$H_{\text{MPL}} = \frac{H_s}{2} \sqrt{2 \ln N} = \frac{4}{2}(3.80) = 2(3.80) = 7.6 \text{ m.}$$

6. The closest available option is:

$$H_{\text{max}} \approx 7.4 \text{ m.}$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

Spectral moments and spectral response

M1. Representative period from H_s and first spectral moment

SPECTRAL ANALYSIS - Spectral Moments and Wave Period. A wave system has $H_s = 3.5$ m and first spectral moment $m_1 = 0.5$ m² rad/s. What is a representative wave period T ?

Options:

- A. $T = 9.6$ s
- B. $T = 6.8$ s
- C. $T = 14.1$ s

Correct answer: A. $T = 9.6$ s

Formulario / principles used

- Spectral moments:

$$m_n = \int_0^\infty \omega^n S(\omega) d\omega$$

- Significant wave height relation:

$$H_s = 4\sqrt{m_0} \Rightarrow m_0 = \left(\frac{H_s}{4}\right)^2$$

- First mean period:

$$T_1 = 2\pi \frac{m_0}{m_1}$$

Step-by-step solution

1. Compute m_0 from significant wave height:

$$m_0 = \left(\frac{H_s}{4}\right)^2 = \left(\frac{3.5}{4}\right)^2 = (0.875)^2 = 0.766 \text{ m}^2.$$

2. Use the first mean period formula:

$$T_1 = 2\pi \frac{m_0}{m_1}.$$

3. Substitute numerically:

$$T_1 = 2\pi \frac{0.766}{0.5} = \frac{4.81}{0.5} = 9.62 \text{ s}.$$

4. The closest option is:

$$T = 9.6 \text{ s}.$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

M2. Representative mean period variant

WAVE STATISTICS / SPECTRAL MOMENTS. A wave system has $H_s = 4.0$ m and first spectral moment $m_1 = 0.80$ m² rad/s. What is a representative mean period T_1 approximately?

Options:

- A. 6.3 s

- B. 7.9 s
C. 9.4 s
D. 12.6 s

Correct answer: B. 7.9 s

Formulario / principles used

$$m_0 = \left(\frac{H_s}{4} \right)^2$$

$$T_1 = 2\pi \frac{m_0}{m_1}$$

Step-by-step solution

1. Compute m_0 :

$$m_0 = \left(\frac{4.0}{4} \right)^2 = (1.0)^2 = 1.0 \text{ m}^2.$$

2. Compute T_1 :

$$T_1 = 2\pi \frac{1.0}{0.80} = \frac{6.283}{0.80} = 7.85 \text{ s}.$$

3. The closest option is:

$$T_1 \approx 7.9 \text{ s}.$$

Source(s): 41107_mock_exam_with_answers / Block 2 reference PDF

R1. Transfer function magnitude from spectral densities

SPECTRAL ANALYSIS: Information used for spectral calculations of a ship is available. Specifically, it is known that, at the frequency ω_a , the wave energy spectral density is $S(\omega_a) = 2.3 \text{ m}^2/\text{rad}$ and the corresponding pitch energy spectral density is $R(\omega_a) = 0.006 \text{ rad}^2/\text{s}/\text{rad}$. What value, approximately, has the magnitude $|\Phi(\omega_a)|$ of the pitch transfer function at the frequency ω_a ?

Options:

- A. $|\Phi(\omega_a)| = 2.9 \text{ deg/m}$
B. $|\Phi(\omega_a)| = 1.3 \text{ deg/m}$
C. $|\Phi(\omega_a)| = 0.15 \text{ deg/m}$

Correct answer: A. $|\Phi(\omega_a)| = 2.9 \text{ deg/m}$

Formulario / principles used

- Response spectrum relation:

$$R(\omega) = |\Phi(\omega)|^2 S(\omega)$$

- Transfer function magnitude:

$$|\Phi(\omega)| = \sqrt{\frac{R(\omega)}{S(\omega)}}$$

- Convert from rad/m to deg/m:

$$1 \text{ rad} = \frac{180}{\pi} \text{ deg}$$

Step-by-step solution

1. Compute the ratio:

$$\frac{R(\omega_a)}{S(\omega_a)} = \frac{0.006}{2.3} = 0.002609 \text{ rad}^2/\text{m}^2.$$

2. Take the square root:

$$|\Phi(\omega_a)| = \sqrt{0.002609} = 0.0511 \text{ rad/m}.$$

3. Convert to degrees per metre:

$$0.0511 \frac{\text{rad}}{\text{m}} \cdot \frac{180}{\pi} = 2.93 \text{ deg/m}.$$

4. The closest option is:

$$|\Phi(\omega_a)| = 2.9 \text{ deg/m}.$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

R2. Transfer function magnitude - numerical variant

SPECTRAL ANALYSIS. At a frequency ω_a , the wave energy spectral density is $S(\omega_a) = 1.8 \text{ m}^2\text{s}/\text{rad}$ and the corresponding pitch response spectral density is $R(\omega_a) = 0.008 \text{ rad}^2\text{s}/\text{rad}$. What is the magnitude of the pitch transfer function $|\Phi(\omega_a)|$ approximately?

Options:

- A. 0.0667 rad/m (3.8 deg/m)
- B. 0.0383 rad/m (2.2 deg/m)
- C. 0.0067 rad/m (0.38 deg/m)
- D. 0.1067 rad/m (6.1 deg/m)

Correct answer: A. 0.0667 rad/m (3.8 deg/m)

Formulario / principles used

$$R(\omega) = |\Phi(\omega)|^2 S(\omega)$$

$$|\Phi(\omega)| = \sqrt{\frac{R(\omega)}{S(\omega)}}$$

$$|\Phi|_{\text{deg/m}} = |\Phi|_{\text{rad/m}} \frac{180}{\pi}$$

Step-by-step solution

1. Compute the spectral-density ratio:

$$\frac{R}{S} = \frac{0.008}{1.8} = 0.004444.$$

2. Take the square root:

$$|\Phi| = \sqrt{0.004444} = 0.0667 \text{ rad/m.}$$

3. Convert to degrees per metre:

$$|\Phi| = 0.0667 \frac{180}{\pi} = 3.82 \text{ deg/m.}$$

Source(s): 41107_mock_exam_with_answers / Block 2 reference PDF

R3. Response spectral density from wave spectrum and transfer function

SPECTRAL ANALYSIS - Response PSD from Wave Spectrum. At frequency ω_a , wave spectral density $S(\omega_a) = 1.8 \text{ m}^2\text{s/rad}$ and pitch transfer function magnitude $Y(\omega_a) = 0.08 \text{ rad/m}$. What is the pitch energy spectral density $R(\omega_a)$?

Options:

- A. $R(\omega_a) = 0.01 \text{ deg}^2\text{s/rad}$
- B. $R(\omega_a) = 8 \text{ deg}^2\text{s/rad}$
- C. $R(\omega_a) = 38 \text{ deg}^2\text{s/rad}$

Correct answer: C. $R(\omega_a) = 38 \text{ deg}^2\text{s/rad}$

Formulario / principles used

- Response spectral density:

$$R(\omega) = |Y(\omega)|^2 S(\omega)$$

- Convert transfer function from radians to degrees:

$$Y_{\text{deg/m}} = Y_{\text{rad/m}} \frac{180}{\pi}$$

Step-by-step solution

1. Convert Y to degrees per metre:

$$Y_{\text{deg/m}} = 0.08 \frac{180}{\pi} = 4.584 \text{ deg/m.}$$

2. Square the transfer function magnitude:

$$Y^2 = (4.584)^2 = 21.0 \text{ deg}^2/\text{m}^2.$$

3. Multiply by the wave spectral density:

$$R(\omega_a) = Y^2 S(\omega_a) = (21.0)(1.8) = 37.8 \text{ deg}^2\text{s/rad.}$$

4. The closest option is:

$$R(\omega_a) = 38 \text{ deg}^2\text{s/rad.}$$

Source(s): 41107_MCQ_Study_Guide / Block 2 reference PDF

R4. Significant heave amplitude of a small boat

SPECTRAL ANALYSIS: A wave system is described by a Bretschneider wave spectrum with $H_s = 2$ m and $T_p = 10$ s. A small boat of length $L = 8$ m is sailing in the wave system. What is the significant amplitude h_{heave} of the boat's heave motion approximately?

Options:

A. $h_{\text{heave}} = 4$ m

B. $h_{\text{heave}} = 1$ m

C. $h_{\text{heave}} = 2$ m

Correct answer: B. $h_{\text{heave}} = 1$ m

Formulario / principles used

- Deep-water peak wavelength estimate:

$$L_p = \frac{gT_p^2}{2\pi}$$

- For a small boat relative to the wavelength:

$$|\Phi_{\text{heave}}| \approx 1$$

- Significant wave amplitude:

$$\zeta_{a,s} = \frac{H_s}{2}$$

- If $|\Phi_{\text{heave}}| \approx 1$:

$$h_{\text{heave}} \approx \zeta_{a,s} = \frac{H_s}{2}$$

Step-by-step solution

1. Estimate the peak wavelength:

$$L_p = \frac{gT_p^2}{2\pi} = \frac{9.81(10)^2}{2\pi} = \frac{981}{6.283} = 156 \text{ m.}$$

2. Compare the boat length with the peak wavelength:

$$L_{\text{boat}} = 8 \text{ m} \ll L_p = 156 \text{ m.}$$

3. Therefore the small boat approximately follows the wave surface in heave:

$$|\Phi_{\text{heave}}| \approx 1.$$

4. Significant wave amplitude is half the significant wave height:

$$\zeta_{a,s} = \frac{H_s}{2} = \frac{2}{2} = 1 \text{ m.}$$

5. Hence:

$$h_{\text{heave}} \approx 1 \text{ m.}$$

Source(s): MCQ Test 2025 / MCQ Test 2025 copy / Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide

Final formula reminder for this group

Regular wave theory

$$\omega = \frac{2\pi}{T} = 2\pi f, \quad k = \frac{2\pi}{L}, \quad c = \frac{L}{T} = \frac{\omega}{k}$$

$$\omega^2 = gk \tanh(kh)$$

$$L_0 = \frac{gT^2}{2\pi}, \quad c = \frac{gT}{2\pi}, \quad c_g = \frac{c}{2} \quad (\text{deep water})$$

$$a = \frac{H}{2}, \quad U_m = \omega a = \frac{2\pi}{T} \frac{H}{2}$$

$$H_{\max} = 0.142 L \tanh(kh)$$

Encounter frequency

$$\omega_e = \omega - \frac{\omega^2}{g} U \cos \beta$$

$$T_e = \frac{2\pi}{\omega_e}, \quad T = \frac{2\pi}{\omega}$$

$$U \text{ [m/s]} = 0.51444 U \text{ [knots]}$$

Wave statistics

$H_s = H_{1/3}$ = mean height of the highest one-third waves

$$m_0 = \int_0^\infty S(\omega) d\omega, \quad H_s = 4\sqrt{m_0}$$

$$P(Z > h_1) = \exp\left(-\frac{h_1^2}{2m_0}\right)$$

$$H_{\text{MPL}} = \frac{H_s}{2} \sqrt{2 \ln N}, \quad N = \frac{T_{\text{storm}}}{T_z}$$

Spectral moments and response spectra

$$m_n = \int_0^\infty \omega^n S(\omega) d\omega$$

$$m_0 = \left(\frac{H_s}{4}\right)^2, \quad T_1 = 2\pi \frac{m_0}{m_1}$$

$$R(\omega) = |\Phi(\omega)|^2 S(\omega)$$

$$|\Phi(\omega)| = \sqrt{\frac{R(\omega)}{S(\omega)}}$$

$$1 \text{ rad} = \frac{180}{\pi} \text{ deg}$$

41107 Marine and Ocean Engineering

Ships: Motion Dynamics, Hydrostatics, Strength and Resistance

Solved MCQ Question Bank - professional formula notation

Scope and ordering

This PDF collects the reviewed questions belonging to the **Ships** block. It includes **ship motion dynamics, encounter frequency / forward speed, Froude-Krylov assumptions, ship hydrostatics, hydrostatic coefficients, strength of ships** and **ship resistance / model testing**. Exact duplicates across the reviewed MCQ documents were removed. Same pattern with different numerical values or different answer options is kept as a separate variant.

The questions are ordered as follows:

1. Ship motion dynamics, encounter frequency and Response Amplitude Operator (RAO).
2. Ship hydrostatics, stability, trim and hydrostatic coefficients.
3. Strength of ships.
4. Ship resistance, model testing and Froude scaling.

For each question, the structure is:

1. Original question statement and original answer options.
2. Correct answer highlighted immediately after the options.
3. **Formulario / principles used** with professional mathematical notation.
4. **Step-by-step solution** with numerical substitution whenever numerical data are present.

Default constants and notation

$$g = 9.81 \text{ m/s}^2, \quad \rho_{\text{water}} \approx 1025 \text{ kg/m}^3, \quad 1 \text{ knot} = 0.51444 \text{ m/s}$$

$$\omega = \frac{2\pi}{T} = 2\pi f, \quad f = \frac{1}{T} = \frac{\omega}{2\pi}, \quad \text{freshwater: } 1 \text{ m}^3 \approx 1 \text{ tonne}$$

Contents

Ship motion dynamics, encounter frequency and Response Amplitude Operator (RAO)

M1. Pitch transfer-function phase

SHIP MOTION DYNAMICS: Transfer functions of a conventional container ship are available. At a given frequency $\omega = 0.6$ rad/s, the real and imaginary parts of the pitch transfer function is $(\Re, \Im) = (-5.0 \times 10^{-2}, 8.0 \times 10^{-2})$ rad/m for wave heading 150° . What is the phase lag ϕ relative to the wave?

Options:

- A. $\phi = 238^\circ$
- B. $\phi = 58^\circ$
- C. $\phi = 122^\circ$

Correct answer: C. $\phi = 122^\circ$

Formulario / principles used

- Complex transfer function:

$$\bar{\Phi}(\omega) = \Re\{\bar{\Phi}\} + i\Im\{\bar{\Phi}\}$$

- Phase angle:

$$\phi = \arg(\bar{\Phi}) = \text{atan2}(\Im\{\bar{\Phi}\}, \Re\{\bar{\Phi}\})$$

- If $\Re < 0$ and $\Im > 0$, the point lies in quadrant II:

$$\phi = 180^\circ - \tan^{-1}\left(\frac{|\Im|}{|\Re|}\right)$$

Step-by-step solution

- Identify the signs of the complex components:

$$\Re = -5.0 \times 10^{-2} = -0.05 < 0, \quad \Im = 8.0 \times 10^{-2} = 0.08 > 0.$$

- Since $\Re < 0$ and $\Im > 0$, the complex number is in quadrant II.

- Compute the reference angle:

$$\alpha = \tan^{-1}\left(\frac{|\Im|}{|\Re|}\right) = \tan^{-1}\left(\frac{0.08}{0.05}\right) = \tan^{-1}(1.6) = 58^\circ.$$

- Convert to the quadrant-II angle:

$$\phi = 180^\circ - \alpha = 180^\circ - 58^\circ = 122^\circ.$$

- Therefore, the phase lag is:

$$\phi = 122^\circ.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

M2. Pitch RAO phase, quadrant II variation

SHIP MOTION DYNAMICS. Transfer functions of a ship are available. At a given frequency, the real and imaginary parts of the pitch RAO are $(\Re, \Im) = (-4.0 \times 10^{-2}, 3.0 \times 10^{-2})$ rad/m. What is the phase angle ε of the complex RAO, expressed in degrees in the interval $[0, 360)$?

Options:

- A. 36.9°
- B. 143.1°
- C. 216.9°
- D. 323.1°

Correct answer: B. 143.1°

Formulario / principles used

$$\varepsilon = \text{atan2}(\Im\{\Phi\}, \Re\{\Phi\})$$

For quadrant II:

$$\varepsilon = 180^\circ - \tan^{-1}\left(\frac{|\Im|}{|\Re|}\right)$$

Step-by-step solution

1. Insert the given complex components:

$$\Re = -4.0 \times 10^{-2} = -0.04, \quad \Im = 3.0 \times 10^{-2} = 0.03.$$

2. The point is in quadrant II because:

$$\Re < 0, \quad \Im > 0.$$

3. Compute the reference angle:

$$\alpha = \tan^{-1}\left(\frac{0.03}{0.04}\right) = \tan^{-1}(0.75) = 36.9^\circ.$$

4. Convert to quadrant II:

$$\varepsilon = 180^\circ - 36.9^\circ = 143.1^\circ.$$

5. Therefore:

$$\varepsilon = 143.1^\circ.$$

Source(s): 41107_mock_exam_with_answers / Block 4 reference PDF

M3. Heave natural frequency of a floating cylinder

SHIP MOTION DYNAMICS: A vertical circular cylinder floats with its longitudinal axis vertically aligned. The cylinder undergoes free oscillations (pure heave); damping is neglected. The cylinder has diameter $D = 0.3$ m and floats at a draught $h = 1.1$ m measured from the bottom of the cylinder, when at rest in still water. At which frequency ω_N does the cylinder oscillate?

Options:

- A. $\omega_N > 3$ rad/s
- B. $\omega_N = 3$ rad/s

C. $\omega_N < 3 \text{ rad/s}$

Correct answer: **C.** $\omega_N < 3 \text{ rad/s}$

Formulario / principles used

- Heave equation without damping:

$$(m + a_{33})\ddot{z} + C_{33}z = 0$$

- Natural angular frequency:

$$\omega_N = \sqrt{\frac{C_{33}}{m + a_{33}}}$$

- For a vertical cylinder, ignoring added mass:

$$C_{33} = \rho g A_w, \quad m = \rho A_w h$$

$$\omega_{N,0} = \sqrt{\frac{\rho g A_w}{\rho A_w h}} = \sqrt{\frac{g}{h}}$$

- Added mass increases the denominator, so:

$$\omega_N < \omega_{N,0}$$

Step-by-step solution

1. Compute the no-added-mass estimate:

$$\omega_{N,0} = \sqrt{\frac{g}{h}} = \sqrt{\frac{9.81}{1.1}}.$$

2. Calculate the value inside the square root:

$$\frac{9.81}{1.1} = 8.918.$$

3. Take the square root:

$$\omega_{N,0} = \sqrt{8.918} = 2.99 \text{ rad/s}.$$

4. Even without added mass, the result is slightly below 3 rad/s:

$$2.99 \text{ rad/s} < 3 \text{ rad/s}.$$

5. With added mass, the denominator becomes $m + a_{33} > m$, so the frequency becomes even smaller:

$$\omega_N = \sqrt{\frac{C_{33}}{m + a_{33}}} < \sqrt{\frac{C_{33}}{m}}.$$

6. Therefore:

$$\boxed{\omega_N < 3 \text{ rad/s}}.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

M4. Heave natural frequency, changed draught

SHIP MOTION DYNAMICS. A vertical circular cylinder floats with its axis vertical and undergoes free oscillations in pure heave. Damping is neglected. The draught measured from the bottom of the cylinder is $h = 0.80$ m. What is the natural angular frequency approximately?

Options:

- A. 2.5 rad/s
- B. 3.5 rad/s
- C. 4.5 rad/s
- D. 5.5 rad/s

Correct answer: B. 3.5 rad/s

Formulario / principles used

$$\omega_n = \sqrt{\frac{g}{h}}$$

where h is the draught of the vertical floating cylinder.

Step-by-step solution

1. Insert the given draught:

$$\omega_n = \sqrt{\frac{9.81}{0.80}}.$$

2. Compute the ratio:

$$\frac{9.81}{0.80} = 12.2625.$$

3. Take the square root:

$$\omega_n = \sqrt{12.2625} = 3.50 \text{ rad/s}.$$

4. Therefore:

$$\boxed{\omega_n \approx 3.5 \text{ rad/s}}.$$

Source(s): 41107_mock_exam_with_answers / Block 4 reference PDF

M5. Encountered period to true wave period

WAVES: A ship sails with a speed of 18 knots. The ship encounters regular waves at an angle $\beta = 135^\circ$ (bow-quartering waves). On the ship, the encountered wave period is $T_{\text{enc}} = 10$ s. What is the absolute (true) wave period T , as measured by a fixed observer relative to the waves?

Options:

- A. $T = 12.5$ s
- B. $T = 13.0$ s
- C. $T = 13.5$ s

Correct answer: B. $T = 13.0$ s

Formulario / principles used

- Speed conversion:

$$U[\text{m/s}] = 0.51444 U[\text{knots}]$$

- Encounter angular frequency:

$$\omega_e = \frac{2\pi}{T_{\text{enc}}}$$

- Deep-water encounter equation:

$$\omega_e = \omega - \frac{\omega^2 U}{g} \cos \beta$$

- True wave period:

$$T = \frac{2\pi}{\omega}$$

Step-by-step solution

1. Convert ship speed:

$$U = 18(0.51444) = 9.26 \text{ m/s.}$$

2. Calculate encountered angular frequency:

$$\omega_e = \frac{2\pi}{10} = 0.628 \text{ rad/s.}$$

3. Evaluate heading cosine:

$$\cos(135^\circ) = -0.707.$$

4. Substitute into the encounter equation:

$$0.628 = \omega - \frac{\omega^2(9.26)}{9.81}(-0.707).$$

5. Since $\cos(135^\circ) < 0$, the quadratic term is positive:

$$0.628 = \omega + 0.667\omega^2.$$

6. Rearrange:

$$0.667\omega^2 + \omega - 0.628 = 0.$$

7. Solve the quadratic equation and keep the positive root:

$$\omega \approx 0.477 \text{ rad/s.}$$

8. Convert to true wave period:

$$T = \frac{2\pi}{0.477} = 13.18 \text{ s.}$$

9. Closest option:

$$\boxed{T \approx 13.0 \text{ s}}.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

M6. Encountered period in head seas, 20 knots

WAVE THEORY - Encountered Period (Head Seas). A ship travels at 20 knots in head seas ($\beta = 180^\circ$). The waves have frequency $f = 0.1 \text{ Hz}$. What is the encountered period T_e ?

Options:

- A. $T_e = 8.2$ s
 B. $T_e = 6.9$ s
 C. $T_e = 6.0$ s

Correct answer: C. $T_e = 6.0$ s

Formulario / principles used

$$\omega = 2\pi f$$

$$\omega_e = \omega - \frac{\omega^2 U}{g} \cos \beta$$

For head seas, $\beta = 180^\circ$, so $\cos \beta = -1$:

$$\omega_e = \omega + \frac{\omega^2 U}{g}$$

$$T_e = \frac{2\pi}{\omega_e}$$

Step-by-step solution

1. Convert ship speed:

$$U = 20(0.51444) = 10.29 \text{ m/s.}$$

2. Convert true wave frequency to angular frequency:

$$\omega = 2\pi(0.1) = 0.628 \text{ rad/s.}$$

3. Compute the forward-speed correction:

$$\frac{\omega^2 U}{g} = \frac{(0.628)^2(10.29)}{9.81} = 0.414 \text{ rad/s.}$$

4. Add the correction for head seas:

$$\omega_e = 0.628 + 0.414 = 1.042 \text{ rad/s.}$$

5. Convert to encountered period:

$$T_e = \frac{2\pi}{1.042} = 6.03 \text{ s.}$$

6. Therefore:

$$T_e \approx 6.0 \text{ s.}$$

Source(s): 41107_MCQ_Study_Guide / Block 4 reference PDF

M7. Encountered period in head seas, 18 knots

FORWARD SPEED / ENCOUNTERED WAVES. A ship travels at 18 knots in head seas. The true wave frequency is $f = 0.10$ Hz. What is the encountered period T_e approximately?

Options:

- A. 6.3 s

- B. 7.2 s
C. 8.5 s
D. 10.0 s

Correct answer: A. 6.3 s

Formulario / principles used

$$U = 0.51444 U_{\text{knots}}, \quad \omega = 2\pi f$$

$$\omega_e = \omega + \frac{\omega^2 U}{g} \quad \text{for head seas}$$

$$T_e = \frac{2\pi}{\omega_e}$$

Step-by-step solution

1. Convert speed:

$$U = 18(0.51444) = 9.26 \text{ m/s.}$$

2. Convert frequency:

$$\omega = 2\pi(0.10) = 0.628 \text{ rad/s.}$$

3. Compute the head-seas encounter frequency:

$$\omega_e = 0.628 + \frac{(0.628)^2(9.26)}{9.81}.$$

4. Evaluate the correction term:

$$\frac{(0.628)^2(9.26)}{9.81} = 0.373 \text{ rad/s.}$$

5. Thus:

$$\omega_e = 0.628 + 0.373 = 1.001 \text{ rad/s.}$$

6. Convert to encountered period:

$$T_e = \frac{2\pi}{1.001} = 6.28 \text{ s.}$$

7. Closest option:

$$\boxed{T_e \approx 6.3 \text{ s}}.$$

Source(s): 41107_mock_exam_with_answers / Block 4 reference PDF

M8. Froude-Krylov assumption in ship motion theory

FROUDE-KRYLOV ASSUMPTION - Ship Motion Theory. In ship motion dynamics, the Froude-Krylov force is calculated under which assumption?

Options:

- A. Wave diffraction always dominates the exciting force.
B. The presence of the hull has no effect on waves.
C. Wave diffraction and radiation are neglected.

Correct answer: B. The presence of the hull has no effect on waves.

Formulario / principles used

- Total excitation is often understood as:

$$F_{\text{exc}} = F_{\text{FK}} + F_{\text{diff}}$$

- Froude-Krylov force is calculated from the undisturbed incident-wave pressure:

$$F_{\text{FK}} = \int_S p_{\text{incident}} \mathbf{n} dS$$

- Diffraction accounts for how the body modifies the incident wave field.

Step-by-step solution

1. The Froude-Krylov force uses the pressure field from the incoming wave.
2. This pressure field is the one that would exist if the body were not present.
3. Therefore the body/hull is assumed not to disturb the wave field.
4. The effect of the body disturbing the wave field is diffraction, which is not part of the pure Froude-Krylov contribution.
5. Hence the correct statement is:

the presence of the hull has no effect on waves.

Source(s): Test_Exam_all extra MCQ / 41107_MCQ_Study_Guide / Block 4 reference PDF

Ship hydrostatics, stability, trim and hydrostatic coefficients

H1. Draught for transverse metacentre in the waterplane

SHIP HYDROSTATICS: A rectangular, box-shaped barge of length $L = 50$ m and beam $B = 7$ m floats at a uniform draught T . What must be the draught for the transverse metacentre (M_T) to lie in the waterplane?

Options:

A. $T = 2.7$ m

B. $T = 2.9$ m

C. $T = 3.1$ m

Correct answer: B. $T = 2.9$ m

Formulario / principles used For a rectangular box barge:

$$KB = \frac{T}{2}$$

$$I_T = \frac{LB^3}{12}, \quad \nabla = LBT$$

$$BM_T = \frac{I_T}{\nabla} = \frac{LB^3/12}{LBT} = \frac{B^2}{12T}$$

$$KM_T = KB + BM_T = \frac{T}{2} + \frac{B^2}{12T}$$

If M_T lies in the waterplane:

$$KM_T = T$$

Step-by-step solution

1. Apply the condition $KM_T = T$:

$$\frac{T}{2} + \frac{B^2}{12T} = T.$$

2. Move $T/2$ to the right-hand side:

$$\frac{B^2}{12T} = \frac{T}{2}.$$

3. Multiply by $12T$:

$$B^2 = 6T^2.$$

4. Solve for T :

$$T = \frac{B}{\sqrt{6}}.$$

5. Substitute $B = 7$ m:

$$T = \frac{7}{\sqrt{6}} = \frac{7}{2.449} = 2.86 \text{ m.}$$

6. Closest option:

$$\boxed{T = 2.9 \text{ m}}.$$

Source(s): MCQ Test 2025 / Test MC exam_Corr / Test_Exam_all

H2. Draught for transverse metacentre, changed beam

SHIP HYDROSTATICS. A rectangular, box-shaped barge has beam $B = 8.0$ m and floats at uniform draught T . What must the draught be for the transverse metacentre to lie in the waterplane?

Options:

- A. $T = 2.5$ m
- B. $T = 3.3$ m
- C. $T = 4.0$ m
- D. $T = 4.6$ m

Correct answer: B. $T = 3.3$ m

Formulario / principles used Using the same rectangular-barge condition:

$$T = \frac{B}{\sqrt{6}}$$

Step-by-step solution

1. Insert the beam:

$$T = \frac{8.0}{\sqrt{6}}.$$

2. Evaluate:

$$\sqrt{6} = 2.449, \quad T = \frac{8.0}{2.449} = 3.27 \text{ m}.$$

3. Closest option:

$$\boxed{T = 3.3 \text{ m}}.$$

Source(s): 41107_mock_exam_with_answers / Block 5 reference PDF

H3. Metacentric height from righting lever

SHIP HYDROSTATICS - Metacentric Height GM_T from GZ . A ship heels $\phi = 5^\circ$ and has a righting lever $GZ = 0.5$ m. What is the transverse metacentric height GM_T ?

Options:

- A. $GM_T = 4.5$ m
- B. $GM_T = 5.3$ m
- C. $GM_T = 5.7$ m

Correct answer: C. $GM_T = 5.7$ m

Formulario / principles used For small-angle stability:

$$GZ = GM_T \sin \phi$$

Therefore:

$$GM_T = \frac{GZ}{\sin \phi}$$

Step-by-step solution

1. Rearrange the righting lever relation:

$$GM_T = \frac{GZ}{\sin \phi}.$$

2. Substitute the numerical values:

$$GM_T = \frac{0.5}{\sin(5^\circ)}.$$

3. Calculate:

$$\sin(5^\circ) = 0.08716.$$

4. Therefore:

$$GM_T = \frac{0.5}{0.08716} = 5.74 \text{ m}.$$

5. Closest option:

$$\boxed{GM_T = 5.7 \text{ m}}.$$

Source(s): *Test_Exam_all extra MCQ / 41107_MCQ_Study_Guide / Block 5 reference PDF*

H4. Stability of a floating square log

SHIP HYDROSTATICS: A homogeneous log of square cross section, side $b = 1 \text{ m}$, and length $l = 6 \text{ m}$ long floats with its length parallel to the water surface and with its cross-sectional sides horizontally and vertically aligned. The log's density is half that of the water. What happens to the log if it is given a small heeling moment which is instantly applied and released? Stillwater conditions are assumed.

Options:

- A.** The log is stable in its initial equilibrium position and will return to this position after the heeling moment is applied and released.
- B.** The log is in an unstable equilibrium in its initial equilibrium position and will find a new equilibrium position after the heeling moment is applied and released.
- C.** The log remains in the heeled position that it reaches after the heeling moment is applied and released since the log is unconditionally stable.

Correct answer: B. The log is initially unstable and will find a new equilibrium position.

Formulario / principles used

$$GM_T = KB + BM_T - KG$$

For the square log:

$$T = \frac{b}{2}, \quad KB = \frac{T}{2}, \quad BM_T = \frac{b^2}{12T}, \quad KG = \frac{b}{2}$$

Stability rule:

$$GM_T > 0 \Rightarrow \text{stable}, \quad GM_T < 0 \Rightarrow \text{unstable}$$

Step-by-step solution

1. Since the log density is half the water density, half of the square section is submerged:

$$T = \frac{b}{2} = \frac{1}{2} = 0.5 \text{ m.}$$

2. Compute the centre of buoyancy:

$$KB = \frac{T}{2} = \frac{0.5}{2} = 0.25 \text{ m} = \frac{b}{4}.$$

3. Compute the metacentric radius:

$$BM_T = \frac{b^2}{12T} = \frac{1^2}{12(0.5)} = \frac{1}{6} = 0.167 \text{ m} = \frac{b}{6}.$$

4. For a homogeneous square log, the centre of gravity is at the geometric centre:

$$KG = \frac{b}{2} = 0.5 \text{ m.}$$

5. Compute transverse metacentric height:

$$GM_T = KB + BM_T - KG = 0.25 + 0.167 - 0.5 = -0.083 \text{ m.}$$

6. Since:

$$GM_T < 0,$$

the initial upright equilibrium is unstable.

7. Therefore, after being disturbed, the log will move to another equilibrium orientation.

Source(s): Test MC exam_Corr / Block 5 reference PDF

H5. Displacement of a rectangular box barge

SHIP HYDROSTATICS - Equilibrium and Buoyancy. A rectangular box barge ($L = 50 \text{ m}$, $B = 8 \text{ m}$) floats in freshwater ($\rho = 1000 \text{ kg/m}^3$) at draught $T = 4 \text{ m}$. What is the displacement in metric tons?

Options:

- A. 1,000 tons
- B. 1,600 tons
- C. 2,000 tons

Correct answer: B. 1,600 tons

Formulario / principles used

$$\nabla = LBT$$

$$\Delta = \rho \nabla$$

$$1 \text{ metric ton} = 1000 \text{ kg}$$

where ∇ is displaced volume and Δ is displacement mass.

Step-by-step solution

1. Compute displaced volume:

$$\nabla = LBT = (50)(8)(4) = 1600 \text{ m}^3.$$

2. Compute displacement mass:

$$\Delta = \rho \nabla = (1000)(1600) = 1,600,000 \text{ kg}.$$

3. Convert to metric tons:

$$\Delta = \frac{1,600,000}{1000} = 1600 \text{ tons}.$$

4. Therefore:

$$\boxed{\Delta = 1600 \text{ tons}}.$$

Source(s): 41107_MCQ_Study_Guide / Block 5 reference PDF

H6. Block coefficient

HYDROSTATIC COEFFICIENTS. A ship in saltwater has $L = 120 \text{ m}$, $B = 20 \text{ m}$, $T = 8 \text{ m}$ and displacement mass $\Delta = 15000 \text{ t}$. Taking $\rho = 1.025 \text{ t/m}^3$, what is the block coefficient C_B approximately?

Options:

- A. 0.56
- B. 0.68
- C. 0.76
- D. 0.91

Correct answer: C. $C_B = 0.76$

Formulario / principles used

$$\nabla = \frac{\Delta}{\rho}$$

$$C_B = \frac{\nabla}{LBT}$$

where ∇ is displaced volume and LBT is the volume of the circumscribing rectangular block.

Step-by-step solution

1. Convert displacement mass to displaced volume:

$$\nabla = \frac{15000}{1.025} = 14634 \text{ m}^3.$$

2. Compute the rectangular block volume:

$$LBT = (120)(20)(8) = 19200 \text{ m}^3.$$

3. Compute the block coefficient:

$$C_B = \frac{14634}{19200} = 0.762.$$

4. Closest option:

$$C_B \approx 0.76.$$

Source(s): 41107_mock_exam_with_answers / Block 5 reference PDF

H7. Trim due to an added load

SHIP HYDROSTATICS - Trim and Longitudinal Stability. A 100 m barge floats on an even keel. A load of 50 tons is added at 30 m forward of midship. The barge has $GM_L = 60$ m and displacement $\Delta = 2000$ tons. What is the change in trim (bow-down positive)?

Options:

- A. Δ trim = +0.75 m bow-down
- B. Δ trim = -0.75 m stern-down
- C. Δ trim = +1.5 m

Correct answer: A. +0.75 m bow-down

Note on this MCQ. The sign is unambiguous: adding weight forward makes the bow go down. A simple total-trim estimate gives about 1.25 m total trim, while the source answer reports the bow-down change as about +0.75 m. The option to select is therefore the positive bow-down one.

Formulario / principles used

- Trimming moment:

$$M_{\text{trim}} = wd$$

- Approximate total trim:

$$t = \frac{wdL}{\Delta GM_L}$$

- If the load is added forward of midship:

bow-down trim is positive.

Step-by-step solution

1. Compute the trimming moment:

$$M_{\text{trim}} = wd = (50)(30) = 1500 \text{ ton m.}$$

2. Use the simplified trim estimate:

$$t = \frac{wdL}{\Delta GM_L}.$$

3. Substitute the numerical values:

$$t = \frac{(50)(30)(100)}{(2000)(60)}.$$

4. Compute numerator and denominator:

$$(50)(30)(100) = 150000, \quad (2000)(60) = 120000.$$

5. Thus:

$$t = \frac{150000}{120000} = 1.25 \text{ m} \quad \text{total trim estimate.}$$

6. Since the load is placed forward of midship, the bow moves down:

$$\text{sign} = + \quad (\text{bow-down}).$$

7. Among the available answers, the correct intended option is:

$$\boxed{+0.75 \text{ m bow-down}}.$$

Source(s): Block 5 reference PDF / generated study bank

Strength of ships

S1. Neutral axis of a thin-walled barge section

STRENGTH OF SHIPS: The cross section of a barge is shown below. All plate thicknesses are $t = 10$ mm. What is the vertical distance z_n from the keel to the neutral axis?

Options:

- A. $z_n = 2.72$ m
- B. $z_n = 2.81$ m
- C. $z_n = 2.95$ m

Correct answer: B. $z_n = 2.81$ m

Formulario / principles used

- Neutral axis for a thin-walled section:

$$z_n = \frac{\sum_i A_i z_i}{\sum_i A_i}$$

- Plate area:

$$A_i = l_i t_i$$

- If all plate thicknesses are equal ($t_i = t$), then t cancels:

$$z_n = \frac{\sum_i l_i z_i}{\sum_i l_i}$$

Step-by-step solution

- Split the cross-section into thin plate elements.
- Since all plate thicknesses are equal:

$$A_i = l_i t \quad \Rightarrow \quad z_n = \frac{\sum_i l_i z_i}{\sum_i l_i}.$$

- From the sketch, the summed plate area is:

$$\sum_i A_i = 0.52 \text{ m}^2.$$

- The first moment of area about the keel is:

$$\sum_i A_i z_i = 1.46 \text{ m}^3.$$

- Compute the neutral-axis height:

$$z_n = \frac{1.46}{0.52} = 2.81 \text{ m}.$$

- Therefore:

$$z_n = 2.81 \text{ m}.$$

Source(s): Test MC exam_Corr / Test_Exam_all / Block 6 reference PDF

S2. Neutral axis, explicit plate lengths

STRENGTH OF SHIPS. The thin-walled barge cross section shown below has uniform plate thickness $t = 10$ mm. What is the vertical distance z_n from the keel to the neutral axis?

Options:

- A. $z_n = 2.32$ m
- B. $z_n = 2.43$ m
- C. $z_n = 2.57$ m
- D. $z_n = 3.00$ m

Correct answer: B. $z_n = 2.43$ m

Formulario / principles used

$$z_n = \frac{\sum_i A_i z_i}{\sum_i A_i}$$

For uniform thickness:

$$z_n = \frac{\sum_i l_i z_i}{\sum_i l_i}$$

Step-by-step solution

1. The plate lengths are:

$$l_i = (10, 10, 10, 6, 6, 1, 1) \text{ m.}$$

2. The corresponding centroid heights are:

$$z_i = (0, 1, 6, 3, 3, 0.5, 0.5) \text{ m.}$$

3. Compute the numerator:

$$\sum_i l_i z_i = (10)(0) + (10)(1) + (10)(6) + (6)(3) + (6)(3) + (1)(0.5) + (1)(0.5).$$

4. Evaluate:

$$\sum_i l_i z_i = 0 + 10 + 60 + 18 + 18 + 0.5 + 0.5 = 107.$$

5. Compute the denominator:

$$\sum_i l_i = 10 + 10 + 10 + 6 + 6 + 1 + 1 = 44.$$

6. Therefore:

$$z_n = \frac{107}{44} = 2.43 \text{ m.}$$

7. Correct option:

$$\boxed{z_n = 2.43 \text{ m.}}$$

Source(s): 41107_mock_exam_with_answers / Block 6 reference PDF

S3. Distributed load equilibrium, triangular gravel load

STRENGTH OF SHIPS: The diagram below shows the distributed weight on a rectangular, box-shaped barge loaded with gravel. The barge is made from steel and its own weight is uniformly distributed along the length with intensity $0.1p$. The barge's length and breadth are 50 m and 8 m, respectively, and it floats at the draught 4 m in freshwater (no heel and no trim). Approximately, what value has p ?

Options:

- A. $p = 53$ tons
- B. $p = 38$ tons
- C. $p = 45$ tons/m

Correct answer: A. $p = 53$ tons (intended as tons/m in the load diagram)

Formulario / principles used

- Freshwater displacement in tonnes for a box barge:

$$\Delta = LBT$$

- Triangular load resultant:

$$W_{\text{gravel}} = \frac{1}{2}Lp$$

- Uniform self-weight resultant:

$$W_{\text{steel}} = L(0.1p)$$

- Static equilibrium:

$$\Delta = W_{\text{gravel}} + W_{\text{steel}}$$

Step-by-step solution

1. Compute displaced volume:

$$\nabla = LBT = (50)(8)(4) = 1600 \text{ m}^3.$$

2. In freshwater, this corresponds approximately to:

$$\Delta = 1600 \text{ tons.}$$

3. Compute triangular gravel load:

$$W_{\text{gravel}} = \frac{1}{2}Lp = \frac{1}{2}(50)p = 25p.$$

4. Compute steel self-weight:

$$W_{\text{steel}} = L(0.1p) = (50)(0.1p) = 5p.$$

5. Total weight:

$$W_{\text{total}} = 25p + 5p = 30p.$$

6. Apply equilibrium:

$$30p = 1600.$$

7. Solve:

$$p = \frac{1600}{30} = 53.3 \text{ tons/m.}$$

8. Closest/intended option:

$$p \approx 53 \text{ tons/m.}$$

Source(s): *Test_Exam_all / 41107_MCQ_Study_Guide / Block 6 reference PDF*

S4. Distributed load equilibrium, changed dimensions

STRENGTH OF SHIPS. The barge below is loaded with gravel. The gravel load is triangular along the whole length with maximum intensity p at midship. The barge's own weight is uniformly distributed with intensity $0.2p$. The barge length is 40 m, breadth is 6 m, and it floats at draught $T = 3$ m in freshwater. Approximately what value has p ?

Options:

- A. $p = 18$ tons/m
- B. $p = 26$ tons/m
- C. $p = 32$ tons/m
- D. $p = 40$ tons/m

Correct answer: B. $p = 26$ tons/m

Formulario / principles used

$$\begin{aligned}\Delta &= LBT \\ W_{\text{gravel}} &= \frac{1}{2}Lp, & W_{\text{own}} &= L(0.2p) \\ LBT &= \frac{1}{2}Lp + L(0.2p)\end{aligned}$$

Step-by-step solution

1. Compute freshwater displacement mass:

$$\Delta = LBT = (40)(6)(3) = 720 \text{ tons.}$$

2. Compute triangular gravel load:

$$W_{\text{gravel}} = \frac{1}{2}(40)p = 20p.$$

3. Compute barge own weight:

$$W_{\text{own}} = (40)(0.2p) = 8p.$$

4. Total load:

$$W_{\text{total}} = 20p + 8p = 28p.$$

5. Equilibrium:

$$28p = 720.$$

6. Solve for p :

$$p = \frac{720}{28} = 25.7 \text{ tons/m.}$$

7. Closest option:

$$p \approx 26 \text{ tons/m.}$$

Source(s): 41107_mock_exam_with_answers / Block 6 reference PDF

S5. Location of greatest bending stress

STRENGTH OF SHIPS - Greatest Bending Stress Location. A barge cross-section has identical plate thicknesses throughout and has an inner bottom plate creating an asymmetric section. The neutral axis is NOT at midheight. Which location experiences the greatest bending stress?

Options:

- A. The plate at deck level
- B. The plate at keel level
- C. Cannot be determined without material properties

Correct answer: A. The plate at deck level

Formulario / principles used

$$\sigma = \frac{My}{I}$$

where M is bending moment, I is second moment of area, and y is distance from the neutral axis. Largest absolute bending stress:

$$|\sigma|_{\max} \text{ occurs where } |y| \text{ is maximum.}$$

Step-by-step solution

1. Bending stress varies linearly with distance from the neutral axis:

$$\sigma \propto y.$$

2. Material properties are not needed to locate the maximum stress in this homogeneous section.
3. The section is asymmetric, and the neutral axis is closer to the keel than to the deck.
4. Therefore:

$$y_{\text{deck}} > y_{\text{keel}}.$$

5. Since:

$$|\sigma| = \left| \frac{My}{I} \right|,$$

the largest stress occurs at the largest $|y|$.

6. Hence the deck plate experiences the greatest bending stress.

Source(s): 41107_MCQ_Study_Guide / Block 6 reference PDF

S6. Deck bending stress from section modulus relation

SHIP STRENGTH - Second Moment of Area (Section Modulus). A simplified ship cross-section has a deck plate at $z = 10$ m and keel at $z = 0$, neutral axis at $z_n = 4$ m. The second moment of area is $I = 20 \text{ m}^4$. For a sagging moment $M = 500 \text{ MNm}$, what is the deck stress?

Options:

- A. $\sigma_{\text{deck}} = 150 \text{ MPa}$
- B. $\sigma_{\text{deck}} = 100 \text{ MPa}$
- C. $\sigma_{\text{deck}} = 60 \text{ MPa}$

Correct answer: A. $\sigma_{\text{deck}} = 150 \text{ MPa}$

Formulario / principles used

$$\sigma = \frac{My}{I}$$

$$y_{\text{deck}} = z_{\text{deck}} - z_n$$

$$1 \text{ MPa} = 10^6 \text{ Pa}, \quad 1 \text{ MNm} = 10^6 \text{ Nm}$$

Step-by-step solution

1. Find the distance from the neutral axis to the deck:

$$y_{\text{deck}} = z_{\text{deck}} - z_n = 10 - 4 = 6 \text{ m.}$$

2. Convert bending moment:

$$M = 500 \text{ MNm} = 500 \times 10^6 \text{ Nm.}$$

3. Apply bending stress formula:

$$\sigma_{\text{deck}} = \frac{My_{\text{deck}}}{I}.$$

4. Substitute numerical values:

$$\sigma_{\text{deck}} = \frac{(500 \times 10^6)(6)}{20}.$$

5. Evaluate:

$$\sigma_{\text{deck}} = 150 \times 10^6 \text{ Pa.}$$

6. Convert to MPa:

$$150 \times 10^6 \text{ Pa} = 150 \text{ MPa.}$$

7. Therefore:

$$\boxed{\sigma_{\text{deck}} = 150 \text{ MPa}}.$$

Source(s): 41107_MCQ_Study_Guide / Block 6 reference PDF

Ship resistance, model testing and Froude scaling

R1. Similarity condition in towing-tank resistance estimation

SHIP RESISTANCE / MODEL TESTING. In towing-tank resistance estimation for ships with free-surface effects, what similarity must be preserved in practice?

Options:

- A. Identical Froude numbers of model and ship
- B. Identical Reynolds numbers of model and ship
- C. Identical Froude and Reynolds numbers simultaneously
- D. Identical Strouhal numbers only

Correct answer: A. Identical Froude numbers of model and ship

Formulario / principles used

$$Fn = \frac{U}{\sqrt{gL}}$$

$$Re = \frac{UL}{\nu}$$

Free-surface wave-making similarity is governed primarily by Fn . Exact Fn and Re similarity cannot generally be achieved simultaneously in the same fluid.

Step-by-step solution

1. Ship resistance includes wave-making and viscous/frictional components.
2. Wave-making depends strongly on the Froude number:

$$Fn = \frac{U}{\sqrt{gL}}.$$

3. Therefore, model and ship are compared at corresponding speeds such that:

$$Fn_m = Fn_s.$$

4. Reynolds number similarity would require:

$$Re_m = Re_s,$$

but this cannot be achieved at the same time as Froude similarity in normal same-fluid model tests.

5. Therefore the preserved similarity is:

$$Fn_m = Fn_s.$$

Source(s): 41107_mock_exam_with_answers / Block 7 reference PDF

R2. Resistance estimation via model tests

SHIP RESISTANCE: In resistance estimation via model tests, it is important to ensure what?

Options:

- A. Identical Reynolds numbers as well as Froude numbers of the full-scale and model-scale ships.

B. Identical Froude numbers of the full-scale and model-scale ships.

C. Identical Reynolds numbers of the full-scale and model-scale ships.

Correct answer: B. Identical Froude numbers of the full-scale and model-scale ships.

Formulario / principles used

$$Fn = \frac{U}{\sqrt{gL}}, \quad Re = \frac{UL}{\nu}$$

$Fn_m = Fn_s$ is the practical model-test condition for free-surface resistance.

Step-by-step solution

1. For a geometrically scaled model, the model length is much smaller than the ship length.
2. If Froude number is matched, the model speed follows:

$$\frac{U_s}{U_m} = \sqrt{\frac{L_s}{L_m}}.$$

3. This does not also make Reynolds number equal, because:

$$Re = \frac{UL}{\nu}$$

depends on both speed and length.

4. Since wave-making is governed by Froude number, the practical condition is:

$$Fn_s = Fn_m.$$

5. Viscous effects are then corrected separately.

Source(s): Test MC exam_Corr / Block 7 reference PDF

R3. Resistance ratio at corresponding speeds

SHIP RESISTANCE: A ship of mass 5,000 tons, length 120 m, is to be represented by a model 3 m long for resistance estimation. If performance is compared at the corresponding speeds, what will the ratio r of ship to model resistance approximately be?

Options:

A. $r = 6 \times 10^4$

B. $r = 6 \times 10^3$

C. $r = 6 \times 10^5$

Correct answer: A. $r = 6 \times 10^4$

Formulario / principles used

$$\alpha = \frac{L_s}{L_m}$$

At corresponding Froude speeds, force/resistance scales approximately as volume:

$$\frac{R_s}{R_m} \approx \alpha^3$$

Step-by-step solution

1. Compute the geometric scale factor:

$$\alpha = \frac{L_s}{L_m} = \frac{120}{3} = 40.$$

2. At corresponding Froude speeds:

$$\frac{R_s}{R_m} \approx \alpha^3.$$

3. Substitute:

$$\frac{R_s}{R_m} \approx 40^3.$$

4. Evaluate:

$$40^3 = 40 \cdot 40 \cdot 40 = 64000 = 6.4 \times 10^4.$$

5. Closest option:

$$r \approx 6 \times 10^4.$$

Source(s): *Test MC exam_Corr / Test_Exam_all / 41107_MCQ_Study_Guide / Block 7 reference PDF*

R4. Resistance ratio, changed model length

SHIP RESISTANCE. A ship of length 180 m is represented by a model 4.0 m long for resistance estimation. If performance is compared at corresponding speeds, what is the ratio R_s/R_m approximately?

Options:

A. 9.1×10^3

B. 9.1×10^4

C. 9.1×10^5

D. 4.5×10^4

Correct answer: B. 9.1×10^4

Formulario / principles used

$$\alpha = \frac{L_s}{L_m}$$

$$\frac{R_s}{R_m} \approx \alpha^3$$

Step-by-step solution

1. Compute length scale:

$$\alpha = \frac{180}{4.0} = 45.$$

2. Apply resistance scaling:

$$\frac{R_s}{R_m} \approx 45^3.$$

3. Evaluate:

$$45^3 = (45)(45)(45) = 91125.$$

4. Scientific notation:

$$91125 = 9.11 \times 10^4.$$

5. Closest option:

$$\boxed{\frac{R_s}{R_m} \approx 9.1 \times 10^4}.$$

Source(s): 41107_mock_exam_with_answers / Block 7 reference PDF

R5. Model weight for Froude scaling

SHIP RESISTANCE - Model Weight for Froude Scaling. A 5,000-ton ship ($L = 120$ m) is modelled at $L_m = 3.5$ m. What is the approximate weight of the model?

Options:

- A. 75 kg
- B. 125 kg
- C. 175 kg

Correct answer: B. 125 kg

Formulario / principles used

$$\alpha = \frac{L_s}{L_m}$$

Mass and weight scale with volume:

$$\frac{m_s}{m_m} = \alpha^3$$

Therefore:

$$m_m = \frac{m_s}{\alpha^3}$$

Step-by-step solution

1. Convert ship mass:

$$m_s = 5000 \text{ tons} = 5,000,000 \text{ kg.}$$

2. Compute length scale:

$$\alpha = \frac{120}{3.5} = 34.29.$$

3. Compute volume/mass scale:

$$\alpha^3 = (34.29)^3 \approx 40,300.$$

4. Compute model mass:

$$m_m = \frac{5,000,000}{40,300} = 124 \text{ kg.}$$

5. Closest option:

$$\boxed{m_m \approx 125 \text{ kg}}.$$

Source(s): *Test_Exam_all extra MCQ / 41107_MCQ_Study_Guide / Block 7 reference PDF*

General formula sheet for this Ships block

Ship motion dynamics and encounter frequency

$$\begin{aligned}\bar{\Phi}(\omega) &= \Re\{\bar{\Phi}\} + i\Im\{\bar{\Phi}\} \\ |\Phi(\omega)| &= \sqrt{\left(\Re\{\bar{\Phi}\}\right)^2 + \left(\Im\{\bar{\Phi}\}\right)^2} \\ \epsilon &= \text{atan2}\left(\Im\{\bar{\Phi}\}, \Re\{\bar{\Phi}\}\right) \\ \omega_e &= \omega - \frac{\omega^2 U}{g} \cos \beta, \quad T_e = \frac{2\pi}{\omega_e} \\ U[\text{m/s}] &= 0.51444 U[\text{knots}] \\ \omega_N &= \sqrt{\frac{C_{33}}{m + a_{33}}}, \quad \omega_{N,0} = \sqrt{\frac{g}{h}}\end{aligned}$$

Hydrostatics, stability and trim

$$\begin{aligned}\nabla &= LBT, \quad \Delta = \rho \nabla \\ KB &= \frac{T}{2}, \quad I_T = \frac{LB^3}{12}, \quad BM_T = \frac{I_T}{\nabla} = \frac{B^2}{12T} \\ KM_T &= KB + BM_T, \quad GM_T = KM_T - KG \\ GZ &= GM_T \sin \phi, \quad GM_T = \frac{GZ}{\sin \phi} \\ KM_T = T &\Rightarrow T = \frac{B}{\sqrt{6}} \text{ for rectangular barge with } M_T \text{ in waterplane} \\ C_B &= \frac{\nabla}{LBT} = \frac{\Delta/\rho}{LBT} \\ t &\approx \frac{wdL}{\Delta GM_L}\end{aligned}$$

Strength of ships

$$z_n = \frac{\sum_i A_i z_i}{\sum_i A_i}$$

For uniform plate thickness:

$$z_n = \frac{\sum_i l_i z_i}{\sum_i l_i}$$

$$W_{\text{triangular}} = \frac{1}{2} Lp, \quad W_{\text{uniform}} = Lq$$

Static equilibrium: $W_{\text{total}} = B_{\text{buoyancy}}$

$$\sigma = \frac{My}{I}, \quad Z = \frac{I}{y_{\max}}, \quad \sigma_{\max} = \frac{M}{Z}$$

Largest bending stress occurs at the largest distance from the neutral axis:

$$|y| = \max.$$

Ship resistance and model testing

$$Fn = \frac{U}{\sqrt{gL}}, \quad Re = \frac{UL}{\nu}$$

$$R = \frac{1}{2}\rho U^2 C_T S$$

$$\alpha = \frac{L_s}{L_m}$$

$$\frac{U_s}{U_m} = \sqrt{\alpha}, \quad \frac{S_s}{S_m} = \alpha^2, \quad \frac{m_s}{m_m} = \alpha^3$$

$$\frac{R_s}{R_m} \approx \alpha^3$$

For ship resistance model tests with free-surface effects:

$$Fn_s = Fn_m \quad \text{is preserved in practice.}$$